MILITARY SPECIFICATION

MULTIPLEXER, TIME DIVISION DIGITAL TD-1069( )/G

This specification is approved for use by EL, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Time Division Digital Multiplexer (TDDM) TD-1069( )/G which is a data transmission unit providing 12 channels of communications. It contains transmit and receive sections. The transmit section multiplexes up to 12 channels of conditioned diphase or teletype data with an aggregate data rate of up to 30 kilobits per second into a single serial output bit stream. The receive section demultiplexes and regenerates the data simultaneously with the same format as the transmit section.(See 6.7.)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall be considered a superseding requirement.

SPECIFICATIONS

NN-P-71	Pallets, Material Handling, Wood, Stringer Construction, 2-way and 4-way (Partial)
QQ-S-781	Strapping, Steel, and Seals

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in inproving this document should be addressed to US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-5805

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PPP-B-585	Boxes, Wood Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	Strapping, Non-Metallic (and Connectors)
PPP-T-97	Tape, Packaging, Industrial Filament Reinforced

Military

MIL-P-116	Preservation-packaging, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-E-15090	Enamel, Equipment, Light Gray (Formula No. 111)

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MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-C-22992	Connectors, Plugs and Receptacle Electrical, Waterproof, Quick Disconnect, Heavy Duty Type
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packaging
MIL-M-38510	Microcircuits, General Specification for

Electronics Command

*EL-CP0195-0001A	Nuclear Survivability Requirements for Time Division Digital Multiplexer TD-1069( )/G(U)
*EL-CP0138-0002A	TEMPEST Requirements and Tests for Time Division Digital Multiplexer TD-1069( )/G(U)

*Classified Document

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STANDARDS

Federal

FED-STD-356	Commercial Packaging of Supplies and Equipment
FED-STD-595	Colors

Military

MIL-STD-105	Standard Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Load for 40" x 48" Pallets
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461 Notice 4	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-471	Maintainability, Verification/ Demonstration
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1281	Internal Transient Control for Solid State Power Supplies
MIL-STD-1332	Tactical Prime, Precise, and Utility Terminologies for Classification of DOD Mobile Electric Power Engine Generator Set Family
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipments and Facilities

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MIL-STD-1562	List of Standard Microcircuits
MIL-STD-188-100	Common Long Haul and Tactical Communication System Technical Standards

DRAWINGS

Electronics Command

SM-D-780122	Case Assembly Electronic Equipment

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number should be stipulated when requesting copies).

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3. REQUIREMENTS

3.1 First article. When specified by the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed to conform with the requirements of this specification and the contract.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 4.4).

3.4 TD-1069 performance requirements.

3.4.1 TD-1069 general characteristics.

3.4.1.1 Operational characteristics. (See 4.8.4). The TDDM shall be capable of multiplexing and demultiplexing up to 12 separate digital data and/or teletypewriter traffic streams into a serial bit stream whose rate is 32.0 kilobits per second. The inputs may originate from separately clocked conditioned diphase data sources at rates of 600, 1200, 2400, 4800, and 9600 bits per second and from teletypewriter units with speeds up to 150.0 bits per second. The multiplexer section shall be capable of interfacing with a demultiplexer section.

3.4.1.2 Interface characteristics. (See 4.8.17). The TDDM multiplexer section shall be capable of interfacing with a demultiplexer through a High Speed Serial Data Buffer TD-1065( )/G unit via wire line transmission paths which may consist of up to 3 miles of WF-16, 1000 feet of WM-130, or a combination of up to 1000 feet of WM-130 and WF-16 whose total length does not exceed 3 miles. Input and output channel sections shall be capable of interfacing with data/teletypewriter sources and sinks via wire line transmission paths with the same characteristics described in the previous sentence.

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3.4.1.3 Multiplexer interunit operating characteristics. (See 4.8.18). TDDM with its master oscillator set at the nominal +0.003% rate shall operate with another TDDM with its master oscillator set at the nominal -0.003% rate. Channels programmed to process conditioned diphase data sources at 600, 1200, 2400, 4800, and 9600 bits per second, but with no data sources connected in either direction or both directions, shall not alarm. Data sources within the frequency tolerances cited in 3.4.3 shall operate error free through this system.

3.4.1.4 Output waveshape. (See 3.4.7.7 and 3.4.8.2). The waveshape shall be such that the rise and fall times, (defined as the time required for the pulse to go from 10% to 90% of its peak-to-peak value) shall be within 5% to 15% of the unit interval at the applicable modulation rate. Transitions in both directions shall be reasonably equal within the limits specified above and be equally affected by shunt capacitance across the line. Properly shaped waveforms shall exhibit smooth exponential curves and contain no points of inflection prior to obtaining maximum amplitudes. (See Figure 1).

3.4.2 Total input bit rate. (See 3.4.5). The sum of all data/teletypewriter traffic streams applied to the input channels of the TDDM shall not exceed 30 kilobits per second.

3.4.3 Data source tolerances. (See 4.8.18). The TDDM shall be capable of accepting data streams which have ±.005% frequency tolerance.

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3.4.4 Teletypewriter channels. (See 4.8.4.4). For teletypewriter traffic a 1200 bits per second channel shall be accept and produce low level balanced teletypewriter transmission at any speed up to 150 bits per second. The speed of the teletypewriter shall be within ±5% of its nominal value and shall have less than 5% bias distortion. The receive section of the TDDM shall demultiplex the teletypewriter signals with less than 20% mark to space, bias, or end distortion.

3.4.5 Channel data rate programming. (See 3.4.2 and 4.8.19). Twelve thumbwheel switches, one corresponding to each channel, shall be provided on the TDDM for data rate and type programming. The data type and rate corresponding to each thumbwheel switch position shall be clearly displayed on the TDDM. The transmit and receive sections of each channel shall always be programmed to process the same type and rate data. An off position shall be provided for each channel. In the off position the receive channel circuitry shall produce a constant output voltage level. The unit shall process data accurately on the active channel/channels with up to eleven channel cards removed. The TDDM shall assume a channel to be programmed OFF if its respective channel card is not included in the mainframe. A rate overload indicator shall be activated if the aggregate data rate programmed on the switches exceeds 30 kilobits per second. A teletypewriter subscriber shall be considered a 1200 bit per second source as described in 3.4.4. The TDDM shall use the programming information provided by the thumbwheel switch position only upon application of power or when a programming push button located on a printed circuit card or the TDDM chassis is activated. TDDM channel programming shall be inhibited if the combined programmed rates exceeds 30 kilobits per second.

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3.4.6 Channel programming switches. (See 4.13). The programming switches shall be located either on their respective channel card or as a group on the TDDM chassis. If mounted on the chassis, they shall be easily removable either individually or as a group as a subassembly without unsoldering wires.

3.4.7 Channel characteristics.

3.4.7.1 Signals. (See 4.8.4.2 and 4.8.4.4). The TDDM shall be capable of interfacing the following signals when they are applied to the unit:

a. Conditioned diphase modulated data streams at any of the data rates cited in 3.4.1.1. The data sources shall conform to MIL-STD-188-100 requirements for data transmission. The TDDM shall be required to extract timing from these data streams. The conditioned diphase format without waveshaping is shown in Figure 2.

b. Balanced NRZ teletypewriter signals cited in 3.3.4. The teletypewriter sources shall conform to MIL-STD-188-100 requirements for low level teletypewriter traffic transmission.

3.4.7.2 Termination. (See 4.8.4.4). Four wire terminations shall be provided for each channel. Channel input/output leads shall be terminated in three MIL approved MS3114P14-19P 19 pin connectors. Four channels shall be accessed per connector. All channel terminations shall be short circuit protected. The channel pin designations shall be as listed in Table 1.

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TABLE I

PIN #	DESCRIPTION
A	HIGH CHANNEL N IN
B	LOW CHANNEL N IN
C	HIGH CHANNEL N OUT
D	LOW CHANNEL N OUT
J	HIGH CHANNEL (N+1) IN
K	LOW CHANNEL (N+1) IN
L	HIGH CHANNEL (N+1) OUT
M	LOW CHANNEL (N+1) OUT
E	HIGH CHANNEL (N+2) IN
F	LOW CHANNEL (N+2) IN
G	HIGH CHANNEL (N+2) OUT
H	LOW CHANNEL (N+2) OUT
N	HIGH CHANNEL (N+3) IN
P	LOW CHANNEL (N+3) IN
R	HIGH CHANNEL (N+3) OUT
S	LOW CHANNEL (N+3) OUT
T	NC
U	CHASSIS GROUND
V	NC

MIL approved 19 pin connector, MS3114P-19P, pin designations in the above TABLE I: N=1 for channel 1-4 connector
N=5 for channel 5-8 connector
N=9 for channel 9-12 connector

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3.4.7.3 Channel input impedance. (See 4.8.9). The input impedance of a channel shall not be less than 5400 ohms with a nominal value of 6000 ohms.

3.4.7.4 Channel input sensitivity. The TDDM shall be capable of detecting and processing balanced and unbalanced input signals as described below:

3.4.7.4.1 Balanced input signal sensitivity. (See 4.8.4.2). Balanced input levels shall range from 0.5 volts P-P to 12 volts P-P.

3.4.7.4.2 Unbalanced input signal sensitivity. (See 4.8.20). Unbalanced input levels shall range from ±3 volts to ±6 volts with respect to signal ground. Unbalanced signal sources shall not exceed 30 feet in cable length from the TDDM.

3.4.7.5 Channel output signals level. (See 4.8.4.2). The open circuit voltage of a channel for balanced transmission shall be positive and negative 6 ±.5 volts. Ripple within a bit interval shall not exceed 0.5% P-P under normal operating conditions. The balance between the positive and negative voltage shall be within 5% of each other. The maximum short circuit current shall not exceed 0.1 amps.

3.4.7.6 Channel output impedance. (See 4.8.10). The channel output impedance shall not exceed 100 ohms.

3.4.7.7 Channel output waveshape. (See 4.8.5). The channel output bit stream shall be waveshaped in accordance with the parameters outlined in 3.4.1.4 at a 9600 bit per second conditioned diphase modulation rate.

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3.4.8 Multiplexer output.

3.4.8.1 Frequency source. (See 4.8.26). The clock used to output data at a nominal 32.0 kilobit per second rate shall be:

a. Jumper strap selectable between the internal master oscillator and the timing extracted by the demultiplexer section of the TDDM as outlined in 3.4.9.1.

b. Switch selectable between either the internal master oscillator/demultiplexer extracted timing or an external 32.0 kHz frequency source.

3.4.8.1.1 Master oscillator frequency accuracy. (See 4.8.4.1). The master oscillator provided in the TDDM transmit section shall have a frequency accuracy of ±0.001% at 23° ±10°C. The frequency tolerance when operating over the environmental temperature range specified herein (see 4.6.1) shall not exceed ±0.003%.

3.4.8.1.2 External frequency source interface characteristics. (See 4.8.26). The TDDM shall accept an external frequency source with a frequency accuracy of not less than 32.0 kHz +0.001%. The impedance of the source shall be 100 ohms ±10%. The frequency source shall conform to MIL-STD-188-100 requirements for balanced data transmission. The input sensitivity of the line receiver within the TDDM shall not be less than 50 mv.

3.4.8.1.3 Data rate. (See 4.8.21). The output data rate of the TDDM shall be 32.0 kilobits per second with the accuracy stated in paragraph 3.4.8.1.1. A test point (see 3.4.17) shall be provided to ground (provide a logical 0 voltage level) the input signal to the conditioned diphase modulator.

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3.4.8.2 Transmission format. (See 4.8.6). The transmission format of the TDDM output bit stream shall be conditioned diphase. The conditioned diphase format without waveshaping is shown in Figure 2. The output signal shall be balanced. The output bit stream shall be waveshaped in accordance with the parameters outlined in 3.4.1.4 at a fixed output modulation rate of 32.0 kilobits per second.

3.4.8.3 Output impedance. (See 4.8.11). The output impedance of the TDDM shall be 135 ohms ±10%.

3.4.8.4 Output signal level. (See 4.8.4.5). The TDDM shall provide an output bit stream with a transmission voltage level of 3 volts P-P ±10% when terminated in 135 ohms (±10%). The maximum short circuit current shall not exceed 0.1 amperes.

3.4.8.4.1 Longitudinal balance. (See 4.8.12). The longitudinal balance of the output signal pair shall not be less then 40 dB.

3.4.9 Demultiplexer input.

3.4.9.1 Data rate. (See 4.8.4). The TDDM receive section shall be capable of accepting a conditioned diphase modulated input bit stream at 32.0 kilobits per second. The receive section shall be capable of extracting timing from the input signal.

3.4.9.2 Input impedance. (See 4.8.13). The input impedance of the TDDM demultiplexer shall be 135 ohms ±10%.

3.4.9.3 Input signal level. (See 4.8.4.6). The TDDM receive section shall be capable of detecting and processing input group signals whose level may range from 0.1 volts P-P to 12 volts P-P.

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3.4.9.3.1 Longitudinal balance. (See 4.8.14). The longitudinal balance of the input signal pair shall not be less than 40 dB.

3.4.10 Synchronization and acquisition of bit count integrity. (See 4.8.15). The TDDM shall be capable of demultiplexer synchronization and acquisition of bit count integrity on a channel connection within 300 msec 90% of the time following the application of signal to the demultiplexer input with a random error rate of 1 in 10^4 bits.

3.4.10.1 Maintenance of bit count integrity. (See 4.8.22). The TDDM shall be capable of maintaining absolute bit count integrity on a channel basis for a period of 24 hours in a random error environment of 1 error in 10^3 bits at a confidence level of 99%. For the test specified in 4.8.22 the TDDM shall maintain BCI over a continuous 24 hour time interval.

3.4.10.2 Re-synchronization and re-acquisition of bit count integrity. (See 4.8.23). The TDDM shall be capable of re-synchronization and re-acquisition of bit count integrity without operator intervention if either the input signal to the receive section of the TDDM is removed for not less than 5 seconds and then reapplied or the power to the TDDM is interrupted for a period of .5 second or less or a period of 1 minute or more.

3.4.11 Connectors. (See 4.4). Power and general purpose connectors shall be of the quick disconnect type.

3.4.11.1 Group connectors. Trompeter type BJ-76 or equivalent triax connectors shall be provided for the group input/output streams of the TDDM. They shall be mounted on the rear of the equipment.

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3.4.11.2 Channel connectors. Three MIL approved MS31114P14-19P 19 pin connectors per MIL-C-22992 shall be provided for channel interconnection to the TDDM. They shall be mounted on the rear of the equipment.

3.4.11.3 External frequency source connector. One MIL approved type BJ-37 twinax connector shall be provided to receive an external, balanced, 32.0 kHz frequency source. It shall be mounted on the rear of the equipment.

3.4.12 Power supply. (See 4.8.16). The power supply shall be an integral replaceable subassembly of the TDDM and shall provide all DC voltages necessary for the operation of the internal circuits of the equipment. The power supply on/off switch shall be located on the front panel of the power supply. The input power shall not exceed 60 watts. The power supply shall be fastened to the front of the chassis with 5/16" hex head slotted screws. There shall be no minimum wait time before reapplication of power to the TDDM subsequent to an AC power outage of any duration for operation to be restored. The power supply shall not protude through the top, bottom, side or back of the chassis.

3.4.12.1 Primary power. The primary power shall have characteristics as contained in MIL-STD-1332 for Type I, Class 1 and 2, Mode I, II and III generator sets. In addition the equipment shall be designed to operate from power having the following steady state voltage and frequency ranges:

a. Voltage. 115 VAC ± 10%

b. Frequency.

(1) 50 Hertz ± 5%

(2) 60 Hertz ± 5%

(3) 400 Hertz ± 5%

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3.4.12.2 Output. The internal power supply(s) of the equipment shall provide the DC outputs for proper equipment operation and shall be kept to a minimum. The internal supply(s) shall be well regulated consistent with their function in the system. The individual internal supplies shall be fused or current limited to provide protection from short circuits within the equipment. The internal supplies shall indicate no degradation in performance over the entire range of input power and environmental conditions specified herein. The power supply shall comply with MIL-STD-1281.

3.4.12.3 Power supply monitor. When the power supply is properly energized, the power supply shall be in the "ON" state as indicated by an energized green light located on the power supply front panel.

3.4.12.4 Size. The TDDM shall be 8 1/2 inches high and 12 inches deep. The width shall not exceed 17 1/4 inches to be suitable for relay rack mounting.

3.4.14 Weight. The weight of the TDDM shall not exceed 50 pounds. A warning label shall be affixed to any unit weighing more than one man lift capability in conformance with paragraph 5.9.11.3.3 of MIL-STD-1472.

3.4.15 Combination case. The TDDM shall have a combination case with the same form factor in the areas of external size and rack mounting bracket tapped hole locations as the High Speed Serial Data Buffer TD-1065( )/G per Electronics Command Drawing SM-D-780112. An environmental protection cover will not be required if the unit is sealed under normal operating conditions. If an environmental cover is required, it should be fastened to the chassis with 5/16" hex head slotted screws.

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3.4.16 Fault location - test facility. (See 4.8.4, 4.8.8, and 4.8.24). The TDDM shall have an integral test facility for monitoring signals in and out of the TDDM and for monitoring operation of the internal circuits. Auxiliary test equipment shall not be required for normal system fault location. The test facility shall include a facility for indicating the presence of acceptable DC supply voltages, a traffic alarm to indicate a lack of activity commensurate with a 32.0 kilobit per second conditioned diphase signal and a functional alarm to indicate a fault within the multiplexer. The integral test facility shall be designed such that it may be operated without interrupting traffic or interfering with normal system operation. The fault location circuitry shall not require any initialization signals such as receive signal activity to become operational subsequent to application of power to the TDDM. An indicator shall be located on the front of each plug-in card for go-no-go fault location. The fault location circuitry shall analyse the fault conditions it receives from all the printed circuit boards within the TDDM, activate the visual indicator on the board causing the fault, and inhibit the visual indicator on the boards which are malfunctioning due to erroneous signals being received from the faulty board. The equipment shall be capable of being maintained by standard military test equipment.

3.4.16.1 Defective panel. (See 4.8.24). The TDDM's test facility shall permit the operator to locate defective panels without use of external test equipment.

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3.4.17 Test points. (See 4.8.15, 4.8.21 and 4.8.24). Voltages and waveforms which are useful in maintenance of the equipment shall be available at suitably located test jacks readily accessible, but not on the front panel. Additional test jacks as required shall be located on the plug-in subassembly panels so as to be accessible with the use of an extender panel. All test jacks shall be short circuit protected. Only operator type adjustments shall be accessible without the extender panel.

3.4.18 Extender panel. (See 4.8.25 and 4.14). The extender panel shall consist of a printed circuit panel with a milti-pin connector at each end mounted in a metal frame. It shall be constructed for insertion between the chassis and any plug-in circuit panel for maintenance and shall contain test points for all connector pins. The extender panel shall be a separate item and shall not be stored within the TD-1069( )/G case. The normal electrical performance of the unit shall not be affected by having up to one channel board and one common board on extender cards at the same time.

3.4.19 Loop back mode. (See 4.8.7). A switch shall be provided on the front panel of the TDDM to interconnect the 32.0 kilobit per second multiplexer output with the demultiplexer input to perform a multiplexer self-test. A visual indicator shall alert the operator to the fact that the multiplexer is in this mode.

3.4.20 Alarms (visible and audible). (See 4.4, 4.8.4, 4.8.8 and 4.8.24). Visual traffic, functional, and individual printed circuit card indicators as described in 3.4.16 shall be provided in the TDDM. All alarm detection circuits shall be operated within 1 second after power application to the TDDM. With the TDDM functioning normally, pushing the audible alarm silencing button described in 3.4.21 shall sound the alarm and light all the visual indicators until it is released. A time guard of 2 seconds shall prevent intermittent alarm indications. Alarm conditions shall be removed

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automatically when proper operation is restored. Upon application or reapplication of 32.0 kilobits per second data to the demultiplexer input, the receive elastic storage registers shall be reinitialized and the underflow/overflow alarm latches shall be reset automatically. Auditory alarms design shall conform to human engineering criteria of MIL-STD-1472 , paragraph 5.3 through 5.3.6. Visual alarm indicator lights shall conform to criteria of MIL-STD-1472, paragraphs 5.2.5.19 for color coding, 5.2.5.10 for brightness level, and 5.2.7.2 for spacing. A relay with SPDT contacts shall be provided within the TDDM for remoting of an alarm condition through a MIL approved MS3114E8-3P connector. With the alarm relay de-energized, pin A of the connector shall be normally closed, pin B shall be normally open, and pin C shall be the common contact of the relay. The relay shall be activated under normal operating conditions and shall be deactivated after a 2 second time guard when a fault occurs.

3.4.21 Audible signal ON-OFF switch. (See 4.8.8). A push button switch shall be provided on the front of the equipment so that the audible signal can be turned off for each occurrence. The operation of this signal shall be such that if the equipment was to recover from a failure after the audible alarm has been turned off, the audible signal will operate until the switch is returned to the normal position. Operation of the alarm on-off switch shall turn off the audible signal without delay.

3.5 Preconditioning. (See 4.7).

3.5.1 Electrical preconditioning. (See 4.3.2, 4.5 and 4.7.1). Prior to submission to the specified quality conformance inspections, each assembled TD-1069( )/G consisting of chassis, plug-in panel assemblies, and power supply plus spare assemblies defined by contract shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process.

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3.5.2 Bounce preconditioning. (See 4.3.2, 4.5 and 4.7.2). Each assembled TD-1069( )/G shall meet the requirements specified herein, without subsequent processing, after subjection to the bounce preconditiong of 4.7.2.

3.6 Service conditions. (See 4.6). The equipment shall meet the following service conditions.

3.6.1 Temperature. (See 4.6.1).

3.6.1.1 Low. (See 4.6.1.1). During and after testing as specified in 4.6.1.1, the equipment shall meet the requirements specified in 3.4.1.1, 3.4.4, 3.4.7.4, 3.4.7.5, 3.4.8.1.1, 3.4.8.4, and 3.4.9.3.

3.6.1.2 High. (See 4.6.1.2). During and after testing as specified in 4.6.1.2, the equipment shall meet the requirements specified in 3.4.1.1, 3.4.4, 3.4.7.4, 3.4.7.5, 3.4.8.1.1, 3.4.8.4, and 3.4.9.3.

3.6.2 Humidity. (See 4.6.2). During and after testing as specified in 4.6.2, the equipment shall meet the requirements specified in 3.4.1.1, 3.4.4, 3.4.7.4, 3.4.7.5, 3.4.8.1.1, 3.4.8.4, and 3.4.9.3. At the conclusion of this test, the equipment shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts and materials. Chassis joints shall not exhibit white salt deposits in excess of one trace of salt deposits per linear inch, not to exceed 50 percent of the brazed fillet width, exceed 3/16 inch in length nor cover a total area of 0.030 inch.

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3.6.3 Low pressure (altitude). (See 4.6.3). During and after testing as specified in 4.6.3, the equipment shall meet the requirements specified in 3.4.1.1, 3.4.4, 3.4.7.4, 3.4.7.5, 3.4.8.1.1, 3.4.8.4, and 3.4.9.3.

3.6.4 Vibration and shock.

3.6.4.1 Vibration. (See 4.6.4). After testing as specified in 4.6.4, the equipment shall meet the requirements specified in 3.4.1.1, 3.4.4, 3.4.7.4, 3.4.7.5, 3.4.8.1.1, 3.4.8.4, and 3.4.9.3.

3.6.4.2 Shock (basic design). (See 4.6.5). After testing as specified in 4.6.4, the equipment shall meet the requirements specified in 3.4.1.1, 3.4.4, 3.4.7.4, 3.4.7.5, 3.4.8.1.1, 3.4.8.4, and 3.4.9.3.

3.6.4.3 Shock (bench handling). (See 4.6.6). After testing as specified in 4.4.6, the equipment shall meet the requirements specified in 3.4.1.1, 3.4.4, 3.4.7.4, 3.4.7.5, 3.4.8.1.1, 3.4.8.4, and 3.4.9.3.

3.6.4.4 Bounce, loose cargo. (See 4.6.7). After testing as specified in 4.6.7, the equipment shall meet the requirements specified in 3.4.1.1, 3.4.4, 3.4.7.4, 3.4.7.5, 3.4.8.1.1, 3.4.8.4, and 3.4.9.3.

3.6.5 Salt fog. (See 4.6.8). After testing as specified in 4.6.8, the equipment shall meet the requirements specified in 3.4.1.1, 3.4.4, 3.4.7.4, 3.4.7.5, 3.4.8.1.1, 3.4.8.4, and 3.4.9.3 and when examined visually with the aid of a 10 power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surfaces or exfoliation (See 6.4.2) of metal.

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3.6.6 Fungus. (See 4.6.9). After the fungus test specified in 4.6.9, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.4.4) as evidenced by growth colonization (see 6.4.3) (which includes branching (see 6.4.1) and sporulation (see 6.4.5)) on or within each cubic foot or fraction thereof, of equipment assembly volume. Isolated instances of partial tublar germination (see 6.4.6) shall not be included in this evaluation.

3.7 Electromagnetic Interference (EMI). (See 4.10). The TD-1069( )/G shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4:

(a) CE02	(a) CS02	(b) RE02	(c) RS03
CE03	(a) CS06	(b) RE02.1	
(a) CE04			
CE05			

(a) Powerline conducted emission and susceptibility requirements are applicable to both 60Hz and 400Hz power.

(b) Applicable from 14kHz to 1000MHz

(c) Applicable from 10kHz to 1000MHz

3.7.1 Bonds and grounds. (See 4.10.1). Bonding for equipment current path returns and shock hazards shall be so installed that expansion, contraction, or movement incident to normal service use and maintenance will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance coatings from the immediate areas of contact. Upon completion of the bond and verification of its DC resistance the equipment finish shall be restored to its original condition in accordance with MIL-F-14072. The DC resistance of bonds and grounds shall not exceed 2.5 milliohms.

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3.7.2 Cables and connectors. (See 4.10.1). All shielded cables shall have the shield terminated in an approved military connector with an EMI backshell to provide bonding of the shield. The use of shielded cables shall be kept to a minimum by the use of alternative suppression techniques such as pin filters. All panel mounted connectors and components and the panel mating surfaces shall be free of non-conductive finishes and provide bonding and grounding with mating connectors and equipment ground plane. The DC bonding resistance for all connectors and panel mounted components shall not exceed 2.5 milliohms.

3.8 Finish. (See 4.4). The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware such as hinges, screws, etc., and necessary touch up after mounting. The top coats of paint finish on front panels shall be semi-gloss light gray enamel in accordance with MIL-E-15090. The outside of the case shall be semi-gloss olive drab, Color No. X-24087 in accordance with FED-STD-595. Front panel markings shall be in black, color No. 17038 per FED-STD-595.

3.9 Marking. (See 4.4).

3.9.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification. Photographically printed aluminum plates shall not be used for front panel marking.

3.9.2 Visibility. Whenever practicable, parts and components shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.9.3 Serial numbers. The TDDM shall have consecutively assigned serial numbers unless otherwise specified by the contracting officer.

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3.10 Microelectronic integrated circuits. (See 4.4). Microelectronic integrated circuit devices used in this equipment shall be selected from MIL-STD-1562. If not included in MIL-STD-1562, they must be submitted for approval as outlined in the contract data requirements list.

3.11 Interchangeability. (See 4.12). Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454.

3.12 Panel assemblies (electrical test). (See 4.9). All TD-1069( )/G panel assemblies and plug-in power supply shall meet the requirements of their respective Test Requirement (TRS) drawings when tested in accordance with 4.9. TRS drawings shall be supplied in accordance with contract or purchase order.

3.13 Plug-in printed circuit cards. (See 4.13). Plug-in printed circuit cards shall utilize separate guide and keying systems. i.e., connector guide pins shall not be used for keying purposes. The use of adhesives to affix or mount keying pins is prohibited. Cards shall provide an adequate surface other than the card edge, for use in pressing cards into place. The use of special tools for card removal is prohibited. Cards shall be capable of rapid and easy removal and insertion without the need for excessive force to perform a replacement or removal function. A wide tapered opening shall be provided on the front edge of the slides to facilitate card insertion. A slotted flange marked with applicable card identification symbol shall be mounted in front of the tapered slide in order to preclude the necessity for insertion at eye level only. The printed circuit card type identification number (designation) shall be clearly visable without removing it from its location in the unit.

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3.14 Safety engineering. (See 4.13).

3.14.1 Personnel hazards (See 4.13). Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.14.3 Warning labels. (See 4.4). Warning labels shall be provided on the equipment in accordance with ANSI Z35.1-1972.

3.14.4 Edge rounding. (See 4.4). Exposed edges shall be rounded to a minimum radius of 0.04 inch (1mm) and exposed corners to a minimum of 0.5 inch (13 mm).

3.15 Operational test. (See 4.8.4). Each fabricated and tested TDDM TD-1069( )/G shall be subjected to a final operational test in accordance with 4.8.4 after all other inspections and tests are completed to determine that the TD-1069( )/G is operational prior to packaging. The requirements of 3.4.11, 3.4.4, 3.4.7.4, 3.4.7.5,3.4.8.1.1, 3.4.8.4, and 3.4.9.3 shall be met.

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3.16 Reliability (See 4.11). The Time Division Digital Multiplexer TD-1069( )/G shall demonstrate a specified mean-time-between-time failure (θ_0) of 2,500 hours. The test as specified in 4.11 shall be conducted to determine compliance to the reliability requirement above.

3.17 Mean-time-to-repair (MTTR or MCT). (See 4.16). The inherent equipment MTTR shall not exceed 15 minutes at organizational level. In addition, the maximum time-to-repair shall not exceed 60 minutes for 95% of all maintenance actions.

3.18 Nuclear survivability. (See 4.17). The TD-1069( )/G shall withstand the nuclear environment specified in EL-CP0195-0001A, when tested in accordance with 4.17. The equipment shall be energized during exposure. Subsequent to the tests, the equipment shall be operable as demonstrated by the test of 3.4.1.1, 3.4.4, 3.4.7.4, 3.4.7.5, 3.4.8.1.1, 3.4.8.4, and 3.4.9.3.

3.19 Workmanship. (See 4.4). The equipment shall be manufactured and assembled in accordance with the requirements of MIL-P-11268 and requirement 9 of MIL-STD-454.

3.20 TEMPEST. (See 4.18) The TD-1069( )/G shall meet the requirement of EL-CP0138-0002A when tested in accordance with 4.18.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilities for Government Inspections shall be made available by the contractor. The Government reserves the right for authorized technical representatives to witness any contractor performed inspections and tests.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First Article inspection (does not include packaging (see 4.3)).

b. Inspection covered by subsidiary documents (See 4.4).

c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.15).

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4.3 First article. Unless otherwise specified in the contract or purchase order, the First Article Inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish 20 first article units of the complete TD-1069( )/G.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table II and shall be performed in the order specified therein except that all hardwired assemblies shall undergo Inprocess Group A Inspection (see 4.5.1a) prior to electrical and bounce preconditioning.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE II - FIRST ARTICLE INSPECTION

Inspections	Rqt. Para.	Test Para.	UNIT NUMBERS 1	2	3	4	5	6-13	14-15	16-20
1. Inspections covered by subsidiary documents		4.4	Inspection to be performed on all units							
2. Electrical preconditioning	3.5.1	4.7.1	Inspection to be performed on all units							
3. Bounce preconditiong	3.5.2	4.7.2	Inspection to be performed on all units							
4. Group A Inspection	(See Table III)		Inspection to be performed on all units							
5. Group B Inspection	(See Table IV)		Inspection to be performed on all units							
6. Nuclear Surviability	3.18	4.17	1							
7. EMI	3.7	4.10					1			
8. TEMPEST	3.20	4.18					2			
9. Group C Inspection										
Low Temperature	3.6.1.1	4.6.1.1		2						
High Temperature	3.6.1.2	4.6.1.2		3						
Humidity	3.6.2	4.6.2			5					See Note D
Low Pressure (Altitude)	3.6.3	4.6.3		1						
Vibration	3.6.4.1	4.6.4			1					
Shock, Basic Design	3.6.4.2	4.6.5			2					
Shock, Bench Handling	3.6.4.3	4.6.6			3					
Bounce Loose Cargo	3.6.4.4	4.6.7			4					
Salt Fog (See Note C)	3.6.5	4.6.8		4						
Fungus (See Note C)	3.6.6	4.6.9				1				

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TABLE II - FIRST ARTICLE INSPECTION (Cont'd)

Inspections	Rqt. Para.	Test Para.	UNIT NUMBERS 1	2	3	4	5	6-13	14-15	16-20
10. Group D Inspection Reliability (See Note E)	3.16	4.11						1		
11. Interface Characteristics	3.4.1.2	4.8.17							1	
12. Interunit Operating Characteristics and and Data Source Tolerances	3.4.1.3, 3.4.5	4.8.18							2	
13. Channel Programing	3.4.5	4.8.19							3	
14. Unbalanced Signal Channel Sensitivity	3.4.7.4.2	4.8.20							4	
15. Data Rate	3.4.8.1.2	4.8.21							5	
16. Maintenance of Bit Count Integrity	3.4.10.1	4.8.22							6	
17. Re-Synchronization Time	3.4.10.2	4.8.23							7	
18. Fault Location	3.4.16	4.8.24							8	
19. Maintainability	3.17	4.16							9	
20. Safety	3.14	4.13							10	

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TABLE II - FIRST ARTICLE INSPECTION (Cont'd)

NOTES:

A. The inspections 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

B. The numbers in the unit columns in the table covered by Inspections 6 through 20 specify the order of inspections for the indicated unit.

C. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

D. These units, after successful completion of Group A and Group B inspections shall be utilized during the training course. The cost of refurbishing these units shall be the sole responsibility of the contractor.

E. Eight units shall be selected at random from the twenty (20),first article units and subjected to the test of 4.11 Reliability.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Parts, Materials, and Processes; General	3.3
Connectors	3.4.11
Alarms (visual and audible)	3.4.20
Finish	3.8
Marking	3.9, 3.14.3
Microelectronic Integrated Circuits	3.10
Workmanship	3.14.4, 3.19

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.7.1, 4.7.2, and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection. Each unit (and each subassembly and assembly identified in Table III on contract or purchase order) shall be inspected for conformance to the inspections specified in Table III. The order of inspection of the subassemblies, assemblies and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in Table III shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in Table III.

b. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table III (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies from the discrete lots which have passed the above Quality Assurance Verification Inspection shall be assembled into the end equipments which shall then be subjected by the contractor to the preconditioning requirements of 4.7 and a 100% visual/mechanical and electrical inspection in accordance with Table III in that order.

NOTE: Assemblies are not required to under go further visual/mechanical and electrical inspection as part of the contractor's 100% Group A Inspection of the end equipment.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQL's indicated in Table III. Group A electrical inspection shall be performed in any order which is satisfactory to the Government, except that the operational test 4.8.4 shall be performed last (see 6.6).

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TABLE III - GROUP A INSPECTION

Inspection	Rqt. Para.	Ref. Fig.	Insp. Para.	AQL Major	AQL Minor
Visual - Mechanical					
Case and Covers	3.19		4.13	1%	4%
Power Supply	3.19		4.13	1%	4%
Panel Assemblies (Including Spares)	3.13, 3.19		4.13	1%	4%
Assembled TD-1069( )/G	3.19		4.13	1%	4%
Extender Panel	3.19		4.13	1%	4%
Channel Programming Switches	3.4.6		4.13	1%	4%
Electrical Performance					
Assemblies covered by TRS Drawings, including spares	3.12		4.9	1%	*
Extender Panels	3.4.18		4.14	1%	*
Operational Characteristics	3.4.1.1	3	4.8.4	1%	*
Teletypewriter Channels	3.4.4	3d	4.8.4.4	1%	*
Channel Termination	3.4.7.2	3d	4.8.4.4	1%	*
Channel Input Sensitivity	3.4.7.4.1	3b	4.8.4.2	1%	*
Channel Output Voltage	3.4.7.5	3b	4.8.4.2	1%	*
Channel Output Waveshape	3.4.7.7	3b	4.8.5	1%	*
Frequency Source	3.4.8.1	11	4.8.26	1%	*
Master Oscillator Frequency Accuracy	3.4.8.1.1	3a	4.8.4.1	1%	*
External Frequency Source	3.4.8.1.2	11	4.8.2.6	1%	*
Transmission Format	3.4.8.3	3e	4.8.6	1%	*
Mux Output Signal Level	3.4.8.4	3e	4.8.4.5	1%	*
DeMux Input Signal Level	3.4.9.3	3f	4.8.4.6	1%	*
Loop Back Mode	3.4.19	3a	4.8.7	1%	*
Alarms	3.4.20	3a	4.8.8	1%	*
Operational Inspection **	3.15		4.8.4	1%	*

*Defects in all characteristics other than [illegible] mechanical are considered major.

**The operational inspection shall be the last test performed as specified in 6.

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TABLE IV

GROUP B INSPECTION

Inspection	Reqt. Para.	Ref. Para.	Insp. Para.	AQL
Interchangeability Mechanical	3.11		4.12	6.5%
Channel Input Impedance	3.4.7.3	4	4.8.9	4.0%
Channel Output Impedance	3.4.7.6	5	4.8.10	4.0%
Multiplexer Output Impedance	3.4.8.3	5	4.8.11	4.0%
Multiplexer Longitudial Balance	3.4.8.4.1	6	4.8.12	4.0%
Demultiplexer Input Impedance	3.4.9.2	4	4.8.13	4.0%
Demultiplexer Longitudinal Balance	3.4.9.3.1	7	4.8.14	4.0%
Synchronization and Acquisition of bit count integrity	3.4.10	8	4.8.15	4.0%
Power Supply	3.4.12	9	4.8.16	4.0%
Extender Panel Compatibility	3.4.18	3a	4.8.25	4.0%

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4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plan. The Group B AQL shall be as shown in Table IV and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in any order satisfactory to the Government.

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4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every month. The unit shall be subjected to temperature and humidity testing in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month.

4.5.3.1.3 Subgroup 3. For this subgroup two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every third month. One unit shall be subjected to EMI and Shock, Bench Handling testing, and one unit shall be subjected to Vibration, Shock, Basic Design, and Bounce, Loose Cargo Testing with each unit subjected to testing in that order.

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every fourth month. One unit shall be subjected to Low Pressure (Altitude) and Salt Fog testing, in that order. One unit shall be subjected to Fungus testing.

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4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the samples units are resubjected to and pass Group A inspection only.

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TABLE V

GROUP C INSPECTION

Inspection	Rqt. Para.	Ref. Para.	Insp. Para.
Subgroup 1			
High Temperature	3.6.1.2		4.6.1.2
Operational	3.15	3	4.8.4
Low Temperature	3.6.1.1		4.6.1.1
Operational	3.15	3	4.8.4
Humidity	3.6.2		4.6.2
Subgroup 2			
Maintenance of Bit Count Integrity	3.4.10.1	8	4.8.22
TEMPEST	3.20		4.18
Subgroup 3			
Vibration	3.6.4.1		4.6.4
Shock, Basic Design	3.6.4.2		4.6.5
Shock, Bench Handling	3.6.4.3		4.6.6
Bounce, Loose Cargo	3.6.4.4		4.6.7
EMI	3.7		4.10
Subgroup 4			
Low Pressure (Altitude)	3.6.3		4.6.3
Salt Fog	3.6.5		4.6.8(Note 1)
Fungus	3.6.6		4.6.9(Note 1)

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

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4.5.5 Group D inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection. Test conditions, methods, and procedures shall be in accordance with 4.11.

TABLE VI - GROUP D INSPECTION

Inspection	Req. Para.	Insp. Para
Reliability	3.16	4.11

4.5.5.1 Sampling for Group D inspection. Ten units shall be selected at random from the first month's production and subjected to the test of 4.11. For subsequent Group D inspections, ten units shall be selected at random from every third month's production.

4.5.5.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract.

4.5.5.3 Reinspection of conforming Group D samples units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Service condition tests.

4.6.1 Temperature test.

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4.6.1.1 Low temperature. (See 3.6.1.1) The equipment shall be tested in accordance with Test Method 502.1, Procedure I of MIL-STD-810. Storage temperature shall be minus 70 degrees F. Storage Time shall be 24 hours. Operation temperature shall be minus 25 degrees F. After stabilization at the operating temperature, the equipment shall operate without degradation in specified performance within 5 minutes after power application. Performance tests, as specified in 4.8.4, shall be made during testing, as specified in Test Method 502.1. After testing, performance tests, as specified in 4.8.4, shall be made.

4.6.1.2 High temperature. (See 3.6.1.2). The equipment shall be tested in accordance with Test Method 501.1, Procedure II of MIL-STD-810. Operating temperature shall be plus 145 degrees F. Performance tests, as specified in 4.8.4 shall be made during testing, as specified in Test Method 501.1. After testing, performance tests, as specified in 4.8.4 shall be made.

4.6.2 Humidity test. (See 3.6.2). The equipment, with all covers removed shall be tested in accordance with Test Method 507.1, Procedure II of MIL-STD-810. Performance testing as specified in 4.8.4 shall be made during the 24 to 48 hour period and the last 5 hour period of each 48 hour cycle. At the conclusion of testing, performance testing as specified in 4.8.4 shall be made. Failure of the equipment to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Low pressure (altitude) test. (See 3.6.3). The equipment shall be tested in accordance with Test Method 500.1, Procedure I of MIL-STD-810. In Step 2, the rate of altitude change shall not exceed 2000 feet per minute. Performance tests, as specified in 4.8.4, shall be made during testing, as specified in Test Method 500.1. After testing, performance tests, as specified in 4.8.4 shall be made.

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4.6.4 Vibration test. (See 3.6.4.1) The equipment shall be tested in accordance with Test Method 514.2, Procedure VIII of MIL-STD-810. Table 514.2-VI for equipment condition "wheeled vehicles" and Curve V of Figure 514.2-6 shall be used. Performance testing as specified in 4.8.4 shall be made after testing as specified in Test Method 514.2.

4.6.5 Shock (basic design) test. (See 3.6.4.2) The equipment shall be tested in accordance with Test Method 516.2, Procedure I of MIL-STD-810. Performance testing as specified in 4.8.4 shall be made after testing as specified in Test Method 516.2, Figure 516.2-2 (peak value (A) g's-20, nominal duration (D)-11 ms).

4.6.6 Shock (bench handling) test. (See 3.6.4.3) The equipment shall be tested in accordance with Test Method 516.2, Procedure V of MIL-STD-810. Performance testing as specified in 4.8.4 shall be made after testing as specified in Test Method 516.2.

4.6.7 Bounce, loose cargo test. (See 3.6.4.4) The equipment shall be tested in accordance with Test Method 514.2, Procedure XI, Part 2 of MIL-STD-810. Performance testing, as specified in 4.8.4, shall be made after testing.

4.6.8 Salt fog test. (See 3.6.5) The equipment shall be tested in accordance with Test Method 509.1 of MIL-STD-810. At the conclusion of the test the equipment shall be examined for evidence of visible corrosion. Performance testing as specified in 4.8.4 shall be made after testing as specified in Test Method 509.1. Failure of the equipment to meet the requirements of 3.6.5 shall constitute failure of this test.

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4.6.9 Fungus test. (See 3.6.6) The equipment, with all covers removed, shall be tested in accordance with Test Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization (see 6.4.3) on 50 percent or more of the area control item after 14 and 28 days. No cleaning of the equipment together with cases and accessories, is permitted for 72 hours prior to the start of fungus test. Failure of the equipment to meet the requirements of 3.6.6 shall constitute failure of this test.

4.7 Preconditioning. (See 3.5).

4.7.1 Electrical preconditioning inspection (see 3.5.1). Electrical precondition inspection for the equipment and panels (including spare panels) shall meet the requirements of 3.5.1.

4.7.1.1 Test details.

a. The equipment shall be mounted on a vibration table in its normal operating position. Test Level B of MIL-STD-781 shall apply for Temperature and Vibration requirements. The equipment shall be connected as shown in Figure 3a. The equipment shall be turned on and operated (all channels processing data) for 24 hours and then turned off for a maximum of 1 hour followed by a second 24 hour operating period.

b. Prior to the start and at the end of each 24 hour operating period, the equipment shall be subjected to the test specified in 4.8.4.

c. Test data shall be recorded at the end of each operating period.

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d. The chassis, plug-in printed circuit panel assemblies and power supplies shall be considered to have been successfully electrically preconditioned when they have individually passed a cumulative 48 hour operating preconditioning period without a failure during the last 24 hour operating period. If during the last 24 hour operating preconditioning period a plug-in panel assembly, power supply, or chassis fails, it may be replaced with a preconditioned panel, power supply, or chassis and the preconditioning continued for a 24 hour failure-free operating period. The restart of a new 48 hour preconditioning period is not required. A plug-in panel assembly, power supply or chassis failing operating preconditioning shall, after repair, be resubmitted to and pass the entire 48 nour operation preconditioning period prior to final electrical assembly tests by the contractor.

e. The unit chassis shall have no less than 48 hours of operational preconditioning nor more than 168 hours of operational preconditioning (exclusive of production test time) when used as a vehicle for preconditioning power supplies and plug-in assemblies (including all spares).

f. Assemblies failing electrical acceptance tests subsequent to satisfactory completion of electrical preconditioning may be repaired with an electrically preconditioned assembly and electrically retested, accepted and delivered without repeat of electrical preconditioning.

4.7.1.2 Marking of electrically preconditioned units. All electrically preconditioned plug-in panel assemblies, power supplies and chassis shall be physically identified by a preconditioning stamp. The stamp shall be affixed to the front panel near the nameplate of each unit which has successfully completed the preconditioning as outlined herein. The stamp on preconditioned plug-in panel assemblies shall be affixed to the bottom side, and to the bottom outside edge of the power supply subassembly. The stamp shall be a "p" in a circle and shall be marked on the preconditioned unit per the Group III requirements of MIL-M-13231.

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4.7.2 Bounce preconditioning. (See 3.5.2). Each unit, in its non-operating condition, shall be placed in its normal operating position on the table of the package tester Type 1000 SC as made by the L.A.B Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. Each unit of each lot shall be subjected to this preconditioning. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.8 Electrical tests.

4.8.1 Standard conditions. Unless otherwise specified all tests shall be performed at 115 volts +5 percent from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperatures, pressure and humidity per MIL-STD-810 unless otherwise specified.

4.8.2 Standard test equipment. Standard test equipment is indicated in test figures by manufacturer and model number. In all cases, equivalent substitutes may be used with prior approval of the Contracting Officer's Technical Representative. The equipment indicated in the figures is listed in Table VII. Standard test equipment shall be provided by the contractor. Government approval is required prior to use of proposed substitute items.

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4.8.3 Special test equipment. Special test equipment is identified by name in the test figures. Unless otherwise specified, special test equipment shall be provided by the contractor subject to approval of the Government. A brief description of this equipment is as follows:

a. Data Source/Error Detector (DS/ED). Provides a conditional diphase data stream at 600, 1200, 2400, 4600, 9600 bits per second which is MIL-STD-188-100 low level compatible. The error detector accepts conditioned diphase MIL-STD-188-100 low level signals at any of the above mentioned data rates and correlates them with the transmitted data stream. An activity, latching loss of bit count integrity (BCI) and error indicator is provided on the front panel. The BCI and error count may be remoted to provide total counts.

b. Variable attenuator test box. Provided attenuation for the data source or multiplexer during input sensitivity measurements.

c. Link error inserter. Provides a technique for haulting or inserting random error rate signals into the 32.0 kilobit per second demultiplexer input. It is used in conjunction with a noise source to provide a random error rate, and a counter connected to display the error rate.

d. 4-Wire Terminal Box. Provides a method of accessing the transmit and receive signal pairs from the three TDDM channel multipin connectors specified in 3.4.7.2.
The channel pin designations are illustrated in Table I. A continuity test similar to 4.14 shall be performed to insure compliance with Table I.

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e. Remote alarm test box. Provides a remote visual indication of the status of the TDDM as determined by the internal fault location circuitry. With the remote alarms relay energized (continuity between pins B and C of the remote alarm connector) a green indicater shall be lit indicating no fault. With the remote alarm relay deenergized (continuity between pins A and C of the remote alarm connector) a red indicator shall be lit to indicate a fault.

f. External 32KHz XTAL Clock/Driver. Provides a 32KHz ±0.001% clock with the electrical characteristics specified in MIL-STD-188-100 for low level balanced data transmission.

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TABLE VII

STANDARD TEST EQUIPMENT

Designation	Function	Manufacturer
HP-5326	Frequency Counter	Hewlett Packard
HP-5326B	Digital Volt Meter	Hewlett Packard
HP-3310B	Signal Generator	Hewlett Packard
HP-3350B	Oscillator/AC Voltmeter	Hewlett Packard
HP-200 CD	Signal Generator	Hewlett Packard
TEK-545	Oscilloscope	Tektronix
P 6042	Current Probe	Tektronix
2002-3	TTY Data Source	Digitech
GR-1390A	Noise Source	General Radio
GR-1434M	Decade Resistor Box	General Radio
433	Ammeter	Weston
904	Voltmeter	Weston
931	Ammeter	Weston

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4.8.4 Operational tests. Successful completion of the channel rate assignments for 4.8.4.2 through 4.8.4.4 shall be considered compliance with 3.4.1.1. All subparagraphs from 4.8.4.1 through 4.8.4.6 are considered part of the operational test.

a. Connect the equipment per Figure 3a.

b. Program channel 1 for 9600 bit/second data and all other channels OFF.

c. Turn the unit on. Observe the FUNCTIONAL, TRAFFIC, and printed circuit card fault indicators and the AUDIBLE ALARM immediately and after approximately 5 seconds. Using the internal fault isolation circuitry, observe that the power supply voltages are within tolerance.

d. The applicable requirements of 3.4.16 and 3.4.20 shall be met.

4.8.4.1 Master oscillator frequency accuracy.

a. Connect the counter input to a test point provided for measuring the master oscillator frequency.

b. Measure the oscillator frequency.

c. The requirements of 3.4.8.1.1 shall be met.

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4.8.4.2 Channel balanced input signal sensitivity and output signal level.

a. Connect the equipment to the 4-wire terminal box per Figure 3b.

b. Program channel 1, 2, and 3 for 9600 bit/second data and all other channels OFF. Set the data source/error detector for 9600 bit/second operation.

c. Connect the data source through the attenuator to the first transmit channel of the group under test and the error detector to this same receive channel.

d. Set the attenuator for minimum attenuation. Note that the error detector indicates no errors occuring. Increase the attenuation until errors occur and then decrease the attenuation until no errors occur for a test interval of at least one minute. Connect the oscilloscope to measure differential voltage across the output terminals of the attenuator.

e. The requirements of 3.4.7.4.1 shall be met.

f. Connect the oscil oscope to the output of the channel under test.

g. The requirements of 3.4.7.5 for signal level and balance shall be met.

h. Connect the data source through the attenuator to the second transmit channel of the group under test and the error detector to the same receive channel and repeat steps d through g.

i. Connect the data source and error detector to the third channel of the group under test and repeat steps d through g.

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j. Program channels 4, 5, and 6 for 9600 bits/second and all other channels OFF. Repeat steps c through i.

k. Program channels 7, 8, and 9 for 9600 bits/second and all other channels OFF. Repeat steps c through i.

l. Program channels 10. 11. and 12 for 9600 bits/second and all other channels OFF. Repeat steps c through i.

4.8.4.3 Channel rate performance.

a. Connect the equipment to the 4-wire terminal box per figure 3c. The oscilloscope is not connected.

b. Program channel group 1 through 6 for 4800 bit/second data and all other channels OFF. Program the data source/error detection for 4800 bit/second operation.

c. Connect the data source to the first channel input of the group and the error detector to the output of the last channel of the group.

d. Interconnect the output of a channel to the input of the next channel in the group. (i.e. the output of channel 1 to the input of channel 2).

e. Note that no error occur over a 4 minute interval.

f. The requirements of 3.4.1.1 shall be met.

g. Program channel group 7 through 12 for 4800 bit/second data and all other channels OFF. Repeat steps c through f.

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h. Program channel group 1 through 12 for 2400 bit/second data. Program the data source (error detector for 2400 bit/second operation. Repeat steps c through f.

i. Program channel group 1 through 12 for 1200 bit/second data. Program the data source/error detector for 1200 bit/second operation. Repeat steps c through f.

j. Program channel group 1 through 12 for 600 bit/second data. Program the data source/error detector for 600 bit/second operation. Repeat steps c through f.

4.8.4.4 Teletypewriter channels and short circuit current.

a. Connect the equipment to the 4-wire terminal box per figure 3d.

b. Program channels 1 through 12 for TTY data.

c. Connect the Digitech TTY data generator to channel 1.

d. Set the generator to put out a steady mark.

e. Short the channel output terminals together through the ammeter and measure the short circuit current.

f. The requirements of paragraph 3.4.7.5 shall be met.

g. Set the generator to put out a steady space. Repeat steps e and f.

h. Program the data generator for 150 bits/second.

i. Introduce 4% mark distortion in the generator.

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j. Measure the mark distortion, space distortion and end distortion of the receive signal.

k. The requirements of 3.4.4 shall be met.

l. Repeat steps d through k for channels 2 through 12.

4.8.4.5 Multiplexer output signal level.

a. Connect the equipment to the TD-1069( )/G per figure 3e.

b. Program channels 1 through 12 for 2400 bit/second data.

c. Observe the multiplexer data stream on the oscilloscope.

d. The requirements of 3.4.8.4 shall be met.

e. Short the high side to the low side of the MUX OUT jacks.

f. Measure the short circuit current of the multiplexer output using a current probe with the oscilloscope.

g. The requirements of 3.4.8.4 shall be met.

4.8.4.6 Demultiplexer input signal level.

a. Connect the equipment to the 4-wire terminal box per figure 3f.

b. Program channels 1 through 12 for 2400 bit/second data.

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c. Connect the data source to the first channel input and the error detector to the output of channel 12. Set the data source to produce 2400 bit/second data.

d. Interconnect the output of a channel to the input of the next channel. (i.e. the output of channel 1 to the input of channel 2).

e. Insert the attenuator box between the multiplexer output and the demultiplexer input. Set the attenuator for minimum attenuation. Note that no errors occur on the error detector.

f. Increase the attenuation until errors occur and then decrease the attenuation to the point that no errors occur.

g. Observe that no bit errors occur over a 1 minute period.

h. Measure the voltage level at the DEMUX in jack.

i. The requirements of 3.4.9.3 shall be met.

4.8.5 Channel output waveshape.

a. Connect the equipment as shown in figure 3c.

b. Program channels 1 through 12 for 2400 bit/second data.

c. Connect the data source to the first channel input and the error detector to the output of channel 12. Set the data source to produce 2400 bit/second data.

d. Interconnect the output of a channel to the input of the next channel. (i.e. the output of channel 1 to the input of channel 2).

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e. Perform steps f through on channel 1.

f. Observe the channel data stream on the oscilloscope.

g. Measure the data high line side rise and fall time.

h. The requirements of 3.4.7.7 shall be met.

i. Measure the data low line side rise and fall time.

j. The requirements of 3.4.1.4 and 3.4.7.7 shall be met.

k. Repeat steps f through j on channel 2 through 12.

4.8.6 Transmission format and multiplexer output waveshape.

a. Connect the equipment as shown in figure 3e. The current probe is not used.

b. Program channels 1 through 12 for 1200 bit/second data.

c. Observe the multiplexed data stream on the oscilloscope.

d. Measure the data pulse width.

e. The requirements of 3.4.8.2 shall be met.

f. Measure the rise and fall times.

g. The requirements of 3.4.1.4 and 3.4.8.2 shall be met.

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4.8.7 Loop back mode.

a. Connect the equipment as shown in figure 3a.

b. Program channels 1 through 12 for 2400 bit/second operation.

c. Remove the cable connecting the multiplexer output to the demultiplexer input.

d. Note the TRAFFIC and AUDIBLE alarms are activated.

e. Place the loop back switch to ON.

f. Note that the TRAFFIC and AUDIBLE alarms go off.

g. Note that the LOOP BACK visual indicator on the TDDM is on.

h. Place the LOOP BACK switch to OFF.

i. Note that the LOOP BACK visual indicator is off.

j. The requirements of 3.4.19 shall be met.

4.8.8 Fault location and alarms.

a. Connect the equipment as shown in figure 3a.

b. One at a time, each of several representative signals, which are available on test points, shall be shorted to ground. The BITE circuitry shall indicate which module is faulty.

c. Turn the TDDM OFF.

d. Observe that the red indicator on the remote alarm test box is lit.

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e. Turn the TDDM ON and observe that the FUNCTIONAL FAILURE LAMP and AUDIBLE ALARM come on.

f. After 5 seconds the alarm indications shall cease and note that the green indicator on the remote alarm test box is lit.

g. Interrupt the multiplex data stream and at the end of approximately 1 second the TRAFFIC FAILURE LAMP, AUDIBLE ALARM and the red indicator on the remote alarm test box should come on.

h. Upon reinstating the multiplexer data, the alarm indications shall cease and the green indication on the remote alarm test box shall light.

i. Interrupt the multiplex data for less than 1 second and observe no output from the alarm indicators and that the red indicator on the remote alarm test box remains extinguished.

j. Interrupt the multiplex data stream.

k. Silence the audible alarm by depressing the ALARM RESET button. Note that the TRAFFIC FAILURE LAMP stays lit and the red indicator on the remote alarm test box remains lit.

l. Reinstate the data.

m. Silence the audible alarm by depressing the ALARM RESET button. Note that the TRAFFIC FAILURE LAMP extinguishes and that the green indicator on the remote alarm test box lights.

n. Depress the Audible Signal ON-OFF switch. Observe that the TRAFFIC FAILURE, FUNCTIONAL FAILURE, and printed circuit board indicators light and that the AUBIBLE ALARM sounds.

o. Release the Audible Signal ON-OFF switch. Observe that all fault indicators cease.

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p. Rotate the METER SELECT switch to each of its positions. Observe that the meter pointer lies within the appropriate color coded area at each position.

q. Set the thumbwheel selector switches for four channels in the 9600 bit/second position and all other channels OFF.

r. Note that without depressing the programing button the RATE OVERLOAD LAMP lights and the AUDIBLE ALARM sounds.

s. Set one thumbwheel selector switch OFF.

t. Note that all alarms cease.

u. The applicable requirements of 3.4.16, 3.4.20 and 3.4.21 shall be met.

4.8.9 Channel input impedance.

a. Set up the equipment as shown in figure 4.

b. Set the LOOP BACK switch ON.

c. Program channels 1 through 12 to the TTY position.

d. Connect S1 to channel 1 as illustrated in the figure.

e. Set S1 to position 1.

f. Observe the oscilloscope and adjust the signal generator for 3V P-P at 1000 Hz.

g. Set S1 to position 2.

h. Observe the oscilloscope and adjust the resistance box for an indication of 3V P-P.

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i. The resistance box will indicate the input impedance of the channel. The requirements of 3.4.7.3 shall be met.

j. Repeat steps d through h for channels 2 through 12.

4.8.10 Channel output impedance.

a. Set up the equipment shown in figure 5.

b. Program channels 1 through 12 for 2400 bit/second data.

c. Connect the data source to channel 1.

d. Connect the channel under test to S1 as ilustrated in the figure.

e. Set S1 to position 1.

f. Observe the signal amplitude on the oscilloscope.

g. Set S1 to position 2.

h. Adjust the decade resistance box until the signal deviation is half the previously observed amplitude.

i. The resistance box will indicate the output impedance of the channel. The requirements of 3.4.7.6 shall be met.

j. Repeat steps d through i for channels 2 through 12.

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4.8.11 Multiplexer output impedance.

a. Set up the equipment shown in figure 5.

b. Program channels 1 through 12 for 2400 bit/second data.

c. Connect the multiplexer output to S1 as illustrated in the figure.

d. Set S1 to position 1.

e. Measure the P-P amplitude of the output signal.

f. Set S1 to position 2.

g. Adjust the resistance decade box until the output signal is half the amplitude measured in step e.

h. The resistance box will indicate the output impedance of the multiplexer. The requirements of 3.4.8.3 shall be met.

4.8.12 Multiplexer longitudinal balance.

a. Connect the equipment as shown in figure 6.

b. Program channels 1 through 12 for 2400 bit/second data.

c. Set the oscillator for an output frequency of 32.0 KHz and 3 volts P-P.

d. Measure the longitudinal balance at the multiplexer output terminal pair.

e. The requirements of 3.4.8.4.1 shall be met.

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4.8.13 Demultiplexer input impedance.

a. Connect the equipment as shown in figure 4 with S1 connected to the demultiplexer input.

b. Set S1 to position 1.

c. Adjust the signal generator for 3V P-P at 32 KHz.

d. Set S1 to position 2.

e. Adjust the resistance box for an indication of 3V P-P.

f. The resistance box will indicate the demultiplexer input impedance. The requirements of 3.4.9.2 shall be met.

4.8.14 Demultiplexer longitudinal balance.

a. Connect the equipment as shown in figure 7.

b. Program channels 1 through 12 for 2400 bit/second data.

c. Set the signal generator for 3V P-P at 32 KHz.

d. Measure the longitudinal balance.

e. The requirements of 3.4.9.3.1 shall be met.

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4.8.15 Synchronization and acquisition of bit count integrity.

a. Connect the equipment as shown in figure 8.

b. Program channel 1 for 9600 bit/second data, channels 2 and 8 for 4800 bits/second, channels 3 and 12 for 2400 bits/second, channels 4,6, and 11 for 1200 bits/second, and channels 5, 7, 9 and 10 for 600 bits/second data.

c. Set the data source/error detector for 9600 bits/second and connect it to channel 1.

d. Connect a counter (in the time interval mode) to the data source/error detector BCK-ERR JACK after calibration.

e. Connect the 64 KHz clock from a test point in the miltiplexer to the random error inserter.

f. Connect a counter to ramdom error inserter (ERR JACK) and adjust the noise source for 3.2/second as indicated on the counter.

g. Interrupt the data stream momentarity by use of the TST ON/OFF switch.

h. Observe the elapsed time on the counter.

i. Perform steps g and h 100 times.

j. The requirements of 3.4.10 shall be met.

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4.8.16 Power supply.

a. Connect the equipment per figure 9.

b. Program channels 1 through 12 for 2400 bit/second data.

c. Set M1 for 60 Hz and 115 VAC.

d. Measure the input current from A.

e. Measure the +5, +12, and -12 volt supply voltage with the DVM.

f. Measure the ripple on the +5, +12, and -12 volt supply voltage with the oscilloscope.

g. Short the output terminals of channels 1 through 12.

h. Repeat steps d through f.

i. Remove the short from the output of channel 1 and connect the data source / error detector set for a rate of 2400 bits/second.

j. Observe that the TDDM runs error free for a period of 5 minutes.

k. The requirements of 3.4.12 and 3.4.12.2 shall be met.

l. Increase the output of M1 to 126.5 VAC.

m. Repeat steps d through k.

n. Decrease the output of M1 to 103.5 VAC.

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o. Repeat steps d through k.

p. Set M1 for 115 VAC and 57 Hz.

q. Repeat steps d through k.

r. Set M1 for 115 VAC and 63 Hz.

s. Repeat steps d through k.

t. Set M1 for the following voltage levels and frequencies repeating steps d through k per each condition.

VOLTAGE (VAC)	FREQUENCY (Hz)
115	50
126.5	50
103.5	50
115	52.5
115	47.5

u. Change the primary power from M1 to M2.

v. Set M2 for the following voltage levels and frequencies repeating steps d through k for each condition.

VOLTAGE (VAC)	FREQUENCY (Hz)
115	400
126.5	400
103.5	400
115	380
115	420

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w. Disconnet M1 and M2 and connect the input to a regulated 115 VAC source.

x. Interrupt the 115 VAC source without the use of the TDDM power switch for approximately 1 second.

y. Note that the green light comes back on and the TDDM resumes normal error free operation.

z. The requirement of 3.4.12 shall be met.

aa. Short the +5 volt supply at a test point within the TDDM.

ab. The requirements of 3.4.12.2 shall be met.

ac. Short the +12 volt supply at a test point within the TDDM.

ad. The requirements of 3.4.12.2 shall be met.

ae. Short the -12 volt supply at a test point within the TDDM.

af. The requirements of 3.4.12.2 shall be met.

4.8.17 Interface characteristics.

a. Connect the equipment as shown in figure 10a.

b. Program channels 1 and 2 for 9600 bits/second, channels 3 through 10 for 1200 bits/second, and channels 11 and 12 for 600 bits/second.

c. Set the two data source/error detector for 9600 bit/second data.

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d. Interconnect the equipment per table VIII.

e. Observe that no errors occur for at least a 5 minute period on either error detector (correlator).

f. Remove the wires from J-1077 #2, pair #2T and J-1077 #2 pair #2R to TDDM #2.

g. Connect 14840 feet (3 miles less 1000 feet) of WF-16 cable as follows:

1. Gray pair from J-1077 #2, pair #2T to TDDM #2, channel #2R.

2. Brown pair from J-1077 #2, pair #2R to TDDM #2, channel #1T.

h. Repeat step e.

i. Remove the WF-16 field wire.

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TABLE VIII

INTERFACE CHARACTERISTICS INTERCONNECTION

TDDM # 1 TERMINATION	TEST EQPT OR J-1077 #1	TEST EQPT OR J-1077 #2	TDDM #2 TERMINATION
MUX OUTPUT	PAIR #1 T	PAIR #1 T	DEMUX INPUT
DEMUX IN	PAIR #1 R	PAIR #1 R	MUX OUTPUT
CHANNEL #1 R	PAIR #2 T	PAIR #2 T	CHANNEL #2 X
CHANNEL #1 X	DATA SOURCE #1	CORRELATOR #1	CHANNEL #2 R
CHANNEL #2 R	CORRELATOR #2	DATA SOURCE #2	CHANNEL #1 X
CHANNEL #2 X	PAIR #2 R	PAIR #2 R	CHANNEL #1 R
CHANNEL #3 R	NO CONNECTION	*	CHANNEL #4 X
CHANNEL #3 X	600 Hz SOURCE	*	CHANNEL #12 R
CHANNEL #4 R	PAIR #3 T	PAIR #3 T	CHANNEL #5 X
CHANNEL #4 X	PAIR #3 R	PAIR #3 R	CHANNEL #3 R
CHANNEL #5 R	PAIR #4 T	PAIR #4 T	CHANNEL #6 X
CHANNEL #5 X	PAIR #4 R	PAIR #4 R	CHANNEL #4 R
CHANNEL #6 R	PAIR #5 T	PAIR #5 T	CHANNEL #7 X
CHANNEL #6 X	PAIR #5 R	PAIR #5 R	CHANNEL #5 R
CHANNEL #7 R	PAIR #6 T	PAIR #6 T	CHANNEL #8 X
CHANNEL #7 X	PAIR #6 R	PAIR #6 R	CHANNEL #6 R
CHANNEL #8 R	PAIR #7 T	PAIR #7 T	CHANNEL #9 X
CHANNEL #8 X	PAIR #7 R	PAIR #7 R	CHANNEL #7 R
CHANNEL #9 R	PAIR #8 T	PAIR #8 T	CHANNEL #10 X
CHANNEL #9 X	PAIR #8 R	PAIR #8 R	CHANNEL #8 R
CHANNEL #10 R	PAIR #9 T	PAIR #9 T	CHANNEL #11 X
CHANNEL #10 X	PAIR #9 R	PAIR #9 R	CHANNEL #9 R
CHANNEL #11 R	PAIR #10 T	PAIR #10 T	CHANNEL #12 X
CHANNEL #11 X	PAIR #10 R	PAIR #10 R	CHANNEL #10 R
CHANNEL #12 R	PAIR #11 T	PAIR #11 T	CHANNEL #3 X
CHANNEL #12 X	PAIR #11 R	PAIR #11 R	CHANNEL #11 R

FOR TDDM's: R => 188 OUT (FROM TDDM TO DATA SINK)
X => 188 IN (FROM SOURCE TO TDDM)

* Jumper TDDM #2 Channel #12R to Channel #4X.

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j. Reconnect J-1077 #2, pair #2 T to TDDM #2, channel #2R.

k. Reconnect J-1077 #2, pair #2R to TDDM #2, channel # 1T.

l. Remove the wires from J-1077 #2, pair # 1T to TDDM and pair #1 R to TDDM #2.

m. Connect 14840 feet of WF-16 cable as follows:

1. Gray pair from J-1077 #2, pair # 1T to TDDM # DEMUX INPUT.

2. Brown pair from J-1077 #2, pair #1R to TDDM #2 MUX OUTPUT.

n. Repeat step e.

o. Disconnect the WF-16 cable from TDDM #2.

p. Interconnect the WF-16 cable to TDDM #2 through the High Speed Serial Data Buffer TD-1065 (XE-1)/G as shown in figure 10b.

q. Repeat step e.

r. The requirements of 3.4.1.2 shall be met.

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4.8.18 Multiplexer interunit operating characteristics.

a. Connect the equipment as shown in figure 11. The oscilloscope, attenuator and 32 KHz XTAL CLOCK/DRIVER are not connected for this test.

b. Program channels 1 and 2 for 9600 bits/second, channel 3 for 4800 bits/second, channel 4 for 2400 bit/second, channel 5 for 1200 bits/second, channels 6 through 9 for 600 bits/second, and channels 10 through 12 OFF.

c. Connect the counter to TDDM #1's master oscillator test point.

d. Adjust for nominal +0.003%.

e. Connect the counter to TDDM #2's master oscillator test point.

f. Adjust for the nominal -0.003%.

g. With no data source connected to either TDDM, observe that during 30 minutes of operation no alarms occur.

h. Connect data source #1 to TDDM #1, channel 1 XMT and error detector #1 to TDDM #2 channel 1 REC.

i. Connect data source #2 to TDDM #2 channel 1, XMT and error detector #2 to TDDM #1 channel 1 REC.

j. Program both data source for 9600 bit/second operation.

k. Observe that no errors occur on either correlator for a period of 5 minutes.

l. Adjust the oscillator in data source #1 for its nominal +0.005%.

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m. Adjust the oscillator in data sourece #2 for its nominal -0.005%.

n. Repeat step k.

o. Adjust the oscillator in data source #1 for its nominal -0.005%.

p. Adjust the oscillator in data source #2 for its nominal +0.005%.

q. Repeat step k.

r. The requirements of 3.4.1.3 shall be met.

s. Readjust the master oscillator in both TDDM's and data sources to their nominal frequency.

4.8.19 Channel data rate programming.

a. Connect the equipment per figure 3a.

b. Turn the LOOP BACK switch ON.

c. Program channel 1, 2, and 3 for 9600 bits/second data, channel 4 for TTY data, and all other channels OFF.

d. Program the data source/error detector for 9600 bit/second data and connect it to channel 1.

e. Observe that the TDDM process data error free for 30 seconds.

f. Turn the thumbwheel switch associated with channel 2 OFF and the thumbwheel switch associated with channel 6 to the 2400 bit/second position.

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g. Observe that no errors occurred during performance of step f.

h. Return the thumbwheel switch associated with channel 2 to the 9600 bit/second position.

i. Note that the RATE OVERLOAD LIGHT lights and that the AUDIBLE ALARM sounds. Silence the audible alarm.

j. Depress the PROGRAM push button.

k. Observe that while performing steps h through j no errors occur.

l. With the oscilloscope look at the REC pairs for channel 5 through 12.

m. Note that they are at constant voltage levels (i.e. no transitional activity).

n. Program channel cards 5 through 12 for 1200 bit/second data.

o. Turn the power OFF.

p. Remove channel cards 5 through 12.

q. Turn the power ON.

r. Note that after 5 seconds all alarms deactivate and that channel 1 is processing data error free.

s. Turn the power OFF.

t. Remove channel cards 2 through 4.

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u. Turn the power ON.

v. Observe that channel 1 is processing data error free for 1 minute.

w. Turn the power OFF.

x Install channel cards 2 through 12.

y. Set the thumbwheel switches for channels 2 through 12 OFF and channel 1 to process 600 bit/second data.

z. Program the data source/error detector to process 600 bit/second data.

aa. Turn the power ON.

ab. Observe that within 5 seconds all alarms cease.

ac. Observe that the TDDM is processing data error free for 1 minute and that no alarms occur.

ad. The requirements of 3.4.5 shall be met.

4.8.20 Unbalanced input signal sensitivity.

a. Connect the equipment per figure 12. The 32.0 KHz XTAL Clock/Driver is not connected for this test.

b. Turn the LOOP BACK switch ON.

c. Program channels 1, 2, and 3 for 9600 bit/second data and all other channels OFF.

d. Program the data source/error detector for 9600 bit/second data.

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e. Perform steps f through k using channel 1.

f. Connect the data source/error detector and attenuator to the channel under test through 30 feet of WF-16.

g. Adjust the attenuator for a ±6 volt signal deviation.

h. Observe no error over a 1 minute period.

i. Adjust the attenuator for a ±3 volt signal deviation.

j. Observe no error over a 1 minute period.

k. The requirements of 3.4.7.4.2 shall be met.

l. Repeat steps f through k with channels 2 and 3 connected to the data source/error detector one at a time.

m. Program channels 4 through 8 for 2400 bit/second data and all other channels OFF.

n. Program the data source/error detector for 2400 bit/second data.

o. Repeat steps f through k with channels 4 through 8 connected to the data source/error detector one at a time.

p. Program channels 9 through 12 for 600 bit/second data and all other channels OFF.

q. Program the data source/error detector for 600 bit/second data.

r. Repeat steps f through k with channels 9 through 12 connected to the data source/error detector one at a time.

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4.8.21 Data rate.

a. Connect the equipment per figure 3a.

b. Ground the input signal to the conditioned diphase modulator using the test point provided.

c. Connect the frequency counter to the MULTIPLEXER OUTPUT.

d. The requirements of 3.4.8.1.3 shall be met.

4.8.22 Maintenance of bit count integrity.

a. Connect the equipment as shown in figure 8.

b. Program channel 1 for 9600 bit/second data, channels 2 and 8 for 4800 bit/second data, channels 3 and 12 for 24 bit/second data, channels 4, 6, and 11 for 1200 bit/second data, and channels 5, 7, 9, and 10 for 600 bit/second data.

c. Set the data source/error detector for 9600 bit/second data and connect it to channel 1.

d. Connect a 64 KHz clock from a test point in the multiplexer to the random error inserter.

e. Connect a counter to ERRP JACK on the random error inserter and adjust the noise source for 32/second.

f. Connect a counter, set to count events, to the appropriate jack for counting losses of bit count integrity.

g. The requirements of 3.4.10.1 shall be met.

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4.8.23 Re-synchronization and re-acquisition of bit count integrity.

a. Connect the equipment as shown in figure 3a.

b. Program channel 1 for 9600 bit/second data, channels 2 and 8 for 4800 bit/second data, channel 3 and 12 for 2400 bit/second data, channels 4, 6, and 11 for 1200 bit/second data, and channels 5, 7, 9, and 10 for 600 bit/second data.

c. Set the data source/error detector for 9600 bit/second data and connect it to channel 1.

d. Note that the error detector runs error free for 1 minute.

e. Remove the 32.0 kilobit/second data from the demultiplexer section by removing the DEMUX INPUT signal for 5 seconds or more before re-application.

f. Note that the multiplexer re-acquires synchronization and bit count integrity (the error detector will stop counting errors), and that all alarms cease.

g. Disconnect the AC power to the TDDM for approximately 0.5 seconds.

h. Repeat step f.

i. Disconnect the AC power for approximately 2 minutes.

j. Repeat step f.

k. The requirements of 3.4.10.2 shall be met.

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4.8.24 Fault location.

a. Connect the equipment as shown in Figure .

b. Turn the AC power OFF.

c. Turn the LOOP BACK Switch OFF and disconnect the DEMUX INPUT.

d. Turn the power ON.

e. Observe that the TRAFFIC, FUNCTIONAL, and power circuit board indicators light and that the AUDIBLE ALARM sounds for approximately 5 seconds.

f. Observe that the TRAFFIC ALARM and AUDIBLE ALARM remain active.

g. Silence the AUDIBLE ALARM.

h. Connect a square wave function generator to the DEMUX INPUT.

i. Slowly increase the frequency from 10 Hz to 100 KHz noting the range of frequencies over which the TRAFFIC ALARM senses and the FUNCTIONAL ALARM activates.

j. The requirements of 3.4.14 shall be met.

k. Disconnect the square wave generator.

l. Turn the LOOP BACK Switch ON.

m. Note that the FUNCTIONAL ALARM ceases and the AUDIBLE ALARM activates.

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n. Silence the AUDIBLE ALARM.

o. Turn the AC power OFF.

p. Remove one common card and one channel card.

q. Place them on extender cards in their respective position in the card cage. Turn the AC power on.

r. Short the test points and other critical signals on the common card and channel card to ground.

s. Note that the FUNCTIONAL ALARM, AUDIBLE ALARM, and printed circuit card indicator on the board containing the simulated fault are active and that all other indicators are off.

t. The requirements of 3.4.16 shall be met.

u. Repeat steps o through t for the remaining common cards.

4.8.25 Extender panel.

a. Connect the equipment as shown in Figure 3a.

b. Program channel 1 for 9600 bit/second data, channels 2 and 8 for 4800 bit/second data, channels 3 and 12 for 2400 bit/second data, channels 4, 6, and 11 for 1200 bit/second data, and channels 5, 7, 9, and 10 for 600 bit/second data.

c. Turn the LOOP BACK switch ON.

d. Turn the AC power OFF.

e. Choose a common card and a channel card programmed for 2400 bit/second to remove from the TDDM.

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f. Place each of these cards on an extender card and replace them in their respective location within the card cage.

g. Turn the AC power ON.

h. Note that after approximately 5 seconds all alarms cease.

i. Choose the channel corresponding to card on the extender panel.

j. Program the data source/error detector for the rate corresponding to that channel and connect set.

k. Note that the channel operates error free for 1 minute.

l. Repeat steps j through k for another channel programmed at a different rate.

m. Repeat steps e through l for the remaining common cards with a different channel card on the extender card each time.

n. The requirements of 3.4.18 shall be met.

4.8.26 External frequency source interface characteristics.

a. Connect the equipment as shown in Figure 11. The frequency counter is not used for this test.

b. Program TDDM #1 and TDDM #2 to process 9600 bit per second data on channels 1, 2, and 3 with all other channels programed OFF.

c. Program the Data Sources and Error Correlators to process 9600 bit per second data and connect them to channel 1.

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d. Observe that no errors occur on the error correlators.

e. Connect the oscilloscope to trigger from channel A and connect one terminal of the MUX OUT binding posts to this channel.

f. Connect one terminal of the DEMUX IN binding posts to channel B and set the oscilloscope for the chopped mode of operation.

g. Observe that the transitions in the waveform displayed by channel B is moving with respect to the transitions in the waveform displayed by channel A.

h. Turn TDDM #2 OFF.

i. Remove the appropriate printed circuit card and change the position of the jumper straps to slave the multiplexer output to the timing extracted from the conditioned diphase signal received by the demultiplexer.

j. Replace the printed circuit card and turn TDDM #2 ON.

k. Observe that the transitions in the waveform displayed by channel B remains fixed with respect to the transitions in the waveform displayed by channel A.

l. Observe that no errors occur on the error correlations over a 1 minute period.

m. Connect the external 32KHz XTAL clock/driver and attenuator as shown in Figure 11.

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n. With the oscilloscope bridged across the output of the attenuator differentially, adjust the attenuator for a peak voltage reading of 12 volts.

o. Set the switch on TDDM #1 for external clock operation.

p. Observe that no errors occur on the error correlator over a 1 minute period.

q. Slowly increase the attenuation until errors are detected by correlator #1. Decrease the attenuation until no errors occur over a 1 minute period.

r. Record the peak-to-peak voltage from the oscilloscope and divide its value in half.

s. Reposition the jumper straps which were changed in step i in TDDM #2.

t. Set the switch on TDDM #1 for internal clock operation. Disconnect the attenuator and 32KHz clock/driver from TDDM #1.

u. Interchange the positions of the two TDDM's and repeat steps e through t.

v. The requirements of 3.4.8.1 and 3.4.8.1.2 shall be met.

4.9 Electrical subassembly test. (see 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings to be supplied as specified by the contract or purchase order.

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4.10 Electromagnetic Interference Test. (see 3.7) The equipment shall be tested for compliance to the requirements of 3.7. Interconnecting cables used during test shall be similar in type, shielding and length to those used in an operational installation. The test shall be performed with the equipment simulating normal operation.

4.10.1 Bonds, grounds and connectors. (see 3.7.1 and 3.7.2) Prior to the performance of EMI tests, the bonding impedance requirements of 3.7.1 and 3.7.2 shall be verified. Bonding impedance of the following items shall be measured with a Shallcross Meter Model 670A or equal:

a. All connector shells to chassis frame.

b. Power line filter body to ground and chassis frame.

c. All surfaces requiring EMI gaskets.

d. All switches, meters, hinge fasteners, transformers, iron core inductors and all mechanical mating surfaces which become exposed during maintenance and repair.

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4.11 Reliability. (See 3.16)

4.11.1 First article reliability test. To determine compliance with paragraph 3.16 during first article inspection, the equipment shall be tested in accordance with Test Plan XXII, Test Level B of MIL-STD-781B and the approved test procedures required by the contract or purchase order.

4.11.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan IV of MIL-STD-781 and the approved test procedures required by the contract or purchase order.

4.11.3 Reliability test conditions. The detail test conditions procedures and methods for the tests described in 4.11.1 and 4.11.2 shall be in accordance with contractor prepared; Government approved Test Plan for the Time Division Digital Multiplexer TD-1069( )/G.

4.11.3.1 Performance parameters to be measured. Performance testing as specified in 4.8.4 shall be performed on all units subjected to Reliability testing. The operational tests of 4.8.4 shall be performed during the "ON" period once a day. The time period expended in the performance of operational tests shall not exceed the "ON" time.

4.11.3.2 Equipment duty cycle. Turn equipment on and let temperature stabilize, hold for 3 hours ("ON" period), then turn off and let temperature stabilize. This cycle shall continue throughout the test.

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4.11.4 Vibration. During testing, the equipment shall be mounted on a vibration table positioned in its normal operating plane with all covers in place. Vibration shall be in a plane perpendicular to the printed circuit cards. Covers may be removed as necessary during test periods, but must be replaced at the conclusion of the electrical test period.

4.11.5 Failure reporting and analysis. Final determination of the failure relevancy and multiple and pattern failure catagorization shall be the responsibility of the procuring activity.

4.11.6 Failure definition. The following paragraphs shall be used as a criteria for definition of a failure:

a. Definitions. The definitions contained in MIL-STD-471, MIL-STD-781B and MIL-STD-721B will apply except as modified herein.

b. Failure. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below performance limits specified in 4.8.4.

c. Failure categories. Failures will be categorized as relevant or non-relevant failures.

d. Relevant failures. A relevant failure is one that can be attributed after failure analysis to any of the following causes:

(1) Design defects
(2) Manufacturing errors
(3) Workmanship defects
(4) Physical deterioration (such as wearout or fatigue)
(5) Tolerance degradation beyond the limits defined on the test data sheets.
(6) Failure of alarm or test circuitry or indicator lamps to indicate a failure or that affect system operation.

Failures that are not proven to be non-relevant will be considered as relevant.

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e. Non-relevant failures.

(1) Alarm or test circuitry or indicator lamps which give an erroneous failure indication and which do not affect system operation.

(2) Damage resulting from improper installation.

(3) Failure of test instrumentation or monitoring equipment which is external to the unit on test.

(4) Damage resulting from accident or mishandling.

(5) Failure due to technician or operator errors.

(6) False alarms caused by test circuitry failure.

(7) Dependent failures as defined by MIL-STD-781B except that paragraph 5.5.1(2) of MIL-STD-781B shall govern in the event of multiple failures.

4.11.7 Accept-reject criteria. The test duration and acceptable number of failures for Fixed Length Test Plan XXII is given in Table 4 of MIL-STD-781B on page 13. The accept/reject criteria for Test Plan IV can be found in paragraph 4.2.8.4 of MIL-STD-781B.

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4.12 Inspection for dimensional interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.13 Visual and mechanical inspection. (see 3.4.6, 3.13, 3.14 and 3.19) Equipment shall be examined for the defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with those portions of 3.14 (safety engineering) which can be determined visually.

4.14 Continuity test. (See 3.4.18) A dc continuity check shall be made between each end connector and the test points.

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4.15 Quality conformance inspection of packaging. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.16 Maintainability. (See 3.17) The mean corrective maintenance time of 15 minutes at the organizational level and the maximum corrective maintenance of 60 minutes at the 95th percentile of the organizational level of maintenance shall be demonstrated using Test Method 9 of MIL-STD-471. The consumer's risk, β, shall be 10%.

4.17 Nuclear tests (see 3.18). Nuclear tests (blast, E.M.P. initial nuclear radiation) shall be performed on the equipment in accordance with 4.17.1 through 4.17.3 inclusive.

4.17.1 Nuclear air blast test. The equipment shall be placed in a nuclear air blast simulation facility and exposed to the air blast environment as specified in 3.18. The equipment shall meet the operational requirements of Section 3 herein after being subjected to this test. Test facilities for this test are available at:

a. BRL, Aberdeen, Maryland.

b. Blast load generator, Waterways Experimental Station, Vicksburg, Mississippi.

c. U.S. Naval Civil Engineering Laboratory, Port Hueneme, California.

d. Sandia Corp., Albuquerque, New Mexicl.

e. Air Force Weapons Laboratory, Albuquerque, New Mexico.

Calibration certification of listed test facilities or their equivalents are required by the Government, 45 days prior to nuclear test. Verification of the wave shape of the test simulator is required. Calibration and wave shape verification shall be supplied to the Government by the contractor.

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4.17.2 Electromagnetic pulse. The equipment shall be pulsed at threat level in accordance with the procedures of the (C) Nuclear Vulnerability Report, CDRL A006, Appendix C for the Time Division Digital Multiplexer TD-1069( )/G dated April 1975 or equivalent.

4.17.3 Initial nuclear radiation test. One Time Division Digital Multiplexer, TD-1069( )/G will be exposed to a LINAC, FXR and a pulsed or fast burst reactor in such a manner that all electronic piece parts and circuits in the set receive the neutron and gamma doses and rates specified in 3.18, Nuclear Survivability. The equipment shall then perform as specified in Section 3. For the purpose of design (not acceptance testing), electronic piece parts and circuits which have nuclear test data available or whose mathematical analysis has indicated that a level of confidence has been established, need not be subjected to further nuclear design testing. Acceptance of these electronic piece parts, circuits and mathematical analysis will be determined by the procuring agency from data supplied to the Government by the contractor. Typical test facilities for initial nuclear radiation tests are located at:

a. Fast Burst Reactors, White Sands, New Mexico.

b. Sandia Pulsed Reactor, Albuquerque, New Mexico.

c. Diamon Ordnance Radiation Facility, Washington, D.C.

d. Gulf General Atomic, La Jolla, California.

e. Northrop Corp., Hawthorne, California.

f. LINAC, WSMR, White Sands, New Mexico.

g. Aurora, HDL, Adelphia, Maryland.

h. HERMES II, Sandia Labs., Albuquerque, New Mexico.

Calibration certification of listed test facilities or their equivalents are required by the Government 45 days prior to nuclear test. Calibration and wave shape verification shall be supplied to the Government by the contractor.

4.18 TEMPEST. The equipment shall be tested as specified in EL-CP0138-002A.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Time Division Digital Multiplexer TD-1069( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Time Division Digital Multiplexer TD-1069( )/G shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method 1C-1.

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5.1.1.4.2 Time Division Digital Multiplexer TD-1069( )/G. Each Time Division Digital Multiplexer TD-1069( )/G shall be individually unit packed method III as follows: Cushion all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, Type CF, Class weather-resistant, variety SW, Grade W5c, designed to protect the unit from shock and vibration hazards during handling and transit. As an alternate, cushioning may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-P-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservation application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be Level A, B or Commercial as specified (see 6.2b). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

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5.2.1 Level A. A quantity of Time Division Digital Multiplexer TD-1069( )/G, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of Time Division Digital Multiplexers TD-1069( )/G, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament

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tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Time Division Digital Multiplexers TD-1069( )/G, unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with FED-STD-356.

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5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

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6. NOTES

6.1 Intended use. The Time Division Digital Multiplexer TD-1069( )/G operates in shelter assemblages of Army Tactical Communications System (ATACS). The unit provides a 12 channel digital transmission capability for combining 600, 1200, 2400, 4800, 9600 bit per second conditioned diphase modulated data sources and/or up to 150 baud MIL-STD-188-100 low level teletypewriter (counting as a 1200 bit per second source) signals. The rates and sources are switch selectable on a channel basis. The aggregate data rate must not exceed 30 kilobits per second. The output from the multiplexer is a serial 32.0 kilobit per second conditional diphase signal which may be routed to another multiplexer directly or through a High Speed Serial Data Buffer TD-1065( )/G.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, Level B or commercial preservation and packing (see section 5).

c. When rough handling test is required.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required, test tools and running spares.

g. First Article Inspection. (see 3.1)

h. Length of cables (if any are required).

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6.3 Environmental. Environmental pollution prevention [illegible]s are contained in the packaging material specifications cited herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Definitions.

6.4.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.4.2 Exfoliation. Exfoliation is corrosion along the grain boundries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4.3 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.4.4 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.4.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

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6.4.6 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5 Group C inspection. (see paragraph 4.5.3.3) Approval to ship may be withheld at the descretion of the Government pending the decision from the Contracting Officer on the adequacy of corrective action.

6.5.1 Initial shipments. (see paragraph 4.5.3.3) Initial shipments of end items will be contingent only upon successful completion of all Group C tests.

6.6 Location of operational inspection. (see paragraph 4.8.4 and 4.5.1) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.7 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TD-1069W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.8 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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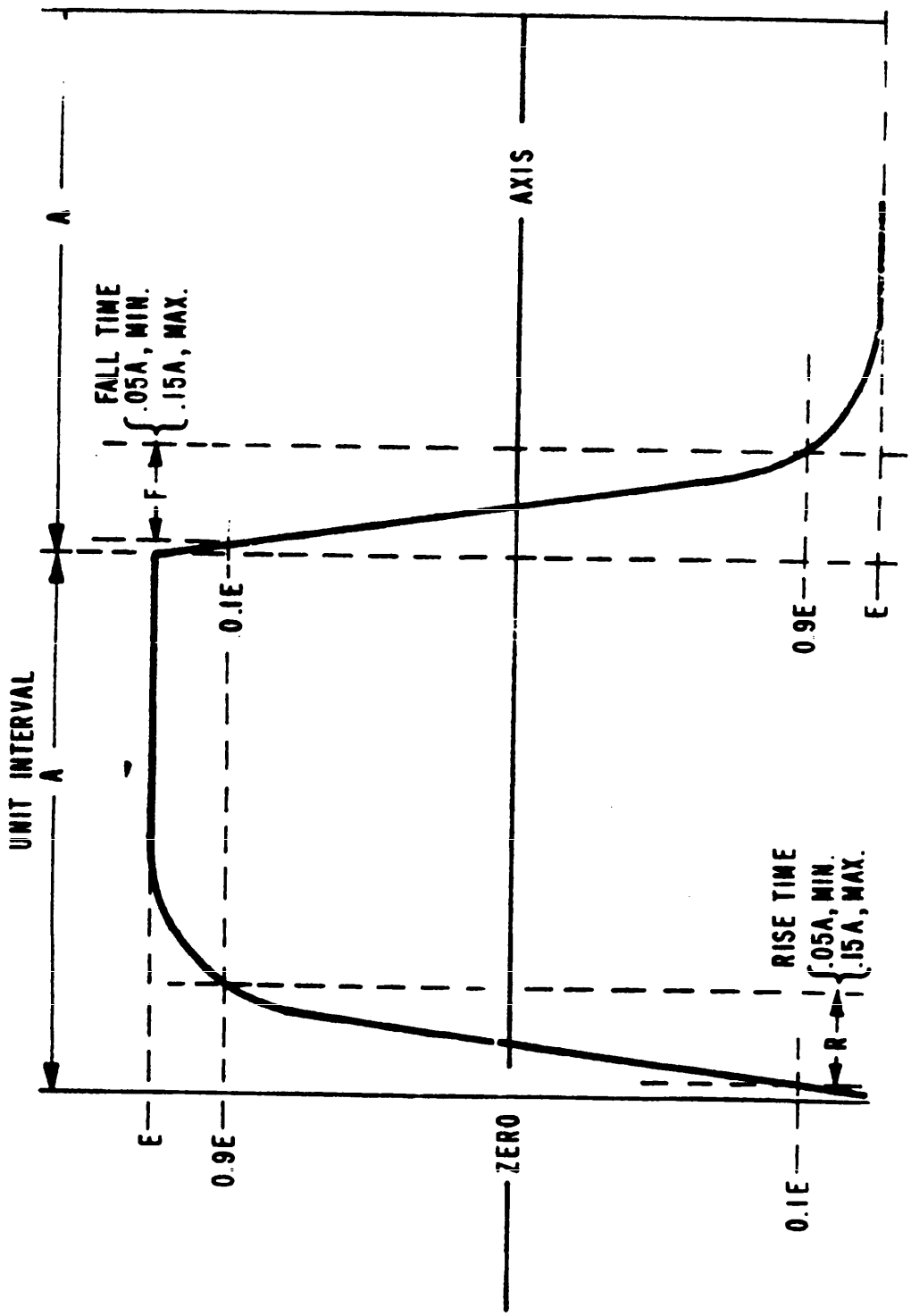


FIG. 1 OUTPUT WAVE SHAPING (CHANNELS AND GROUP)

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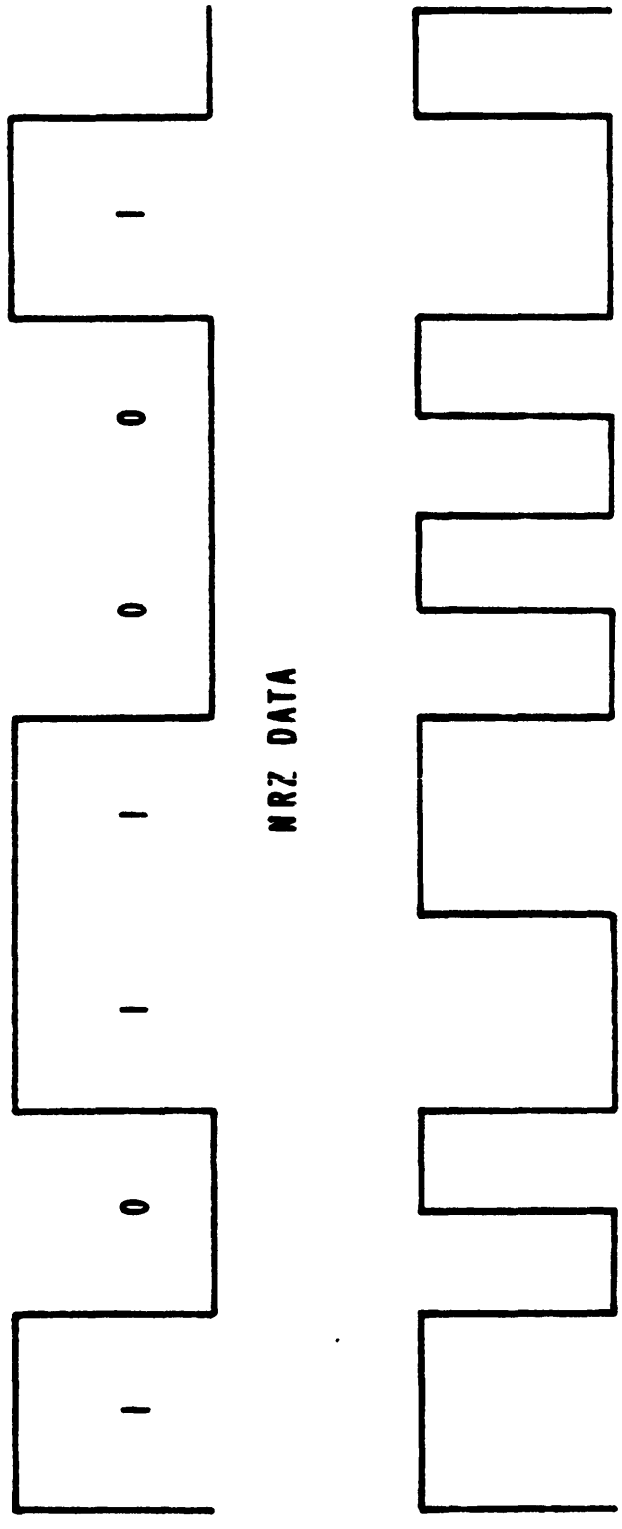


FIG. 2 CONDITIONED DIPHASE MODULATED OUTPUT

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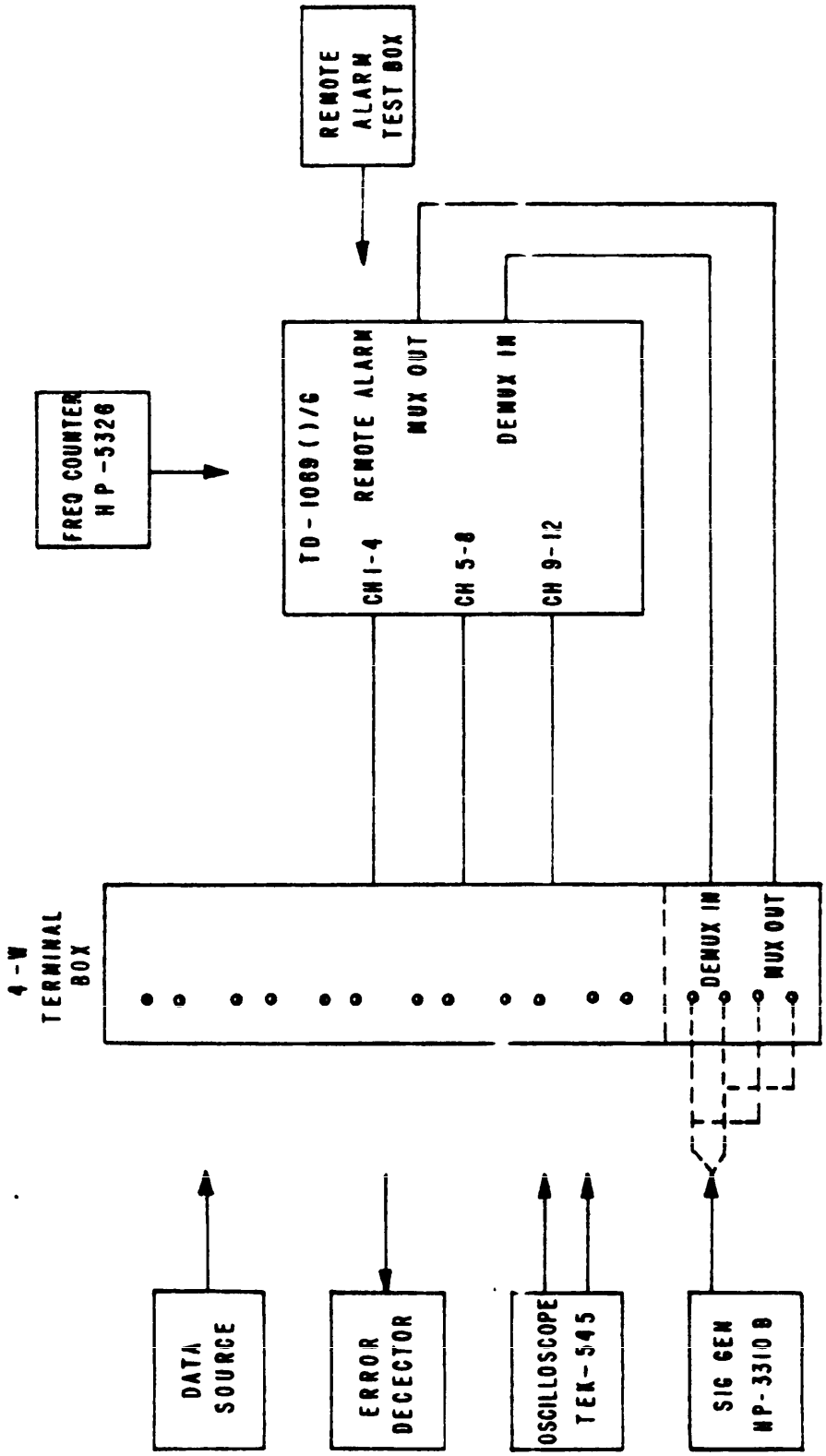


FIG. 3a GENERAL TEST SET-UP

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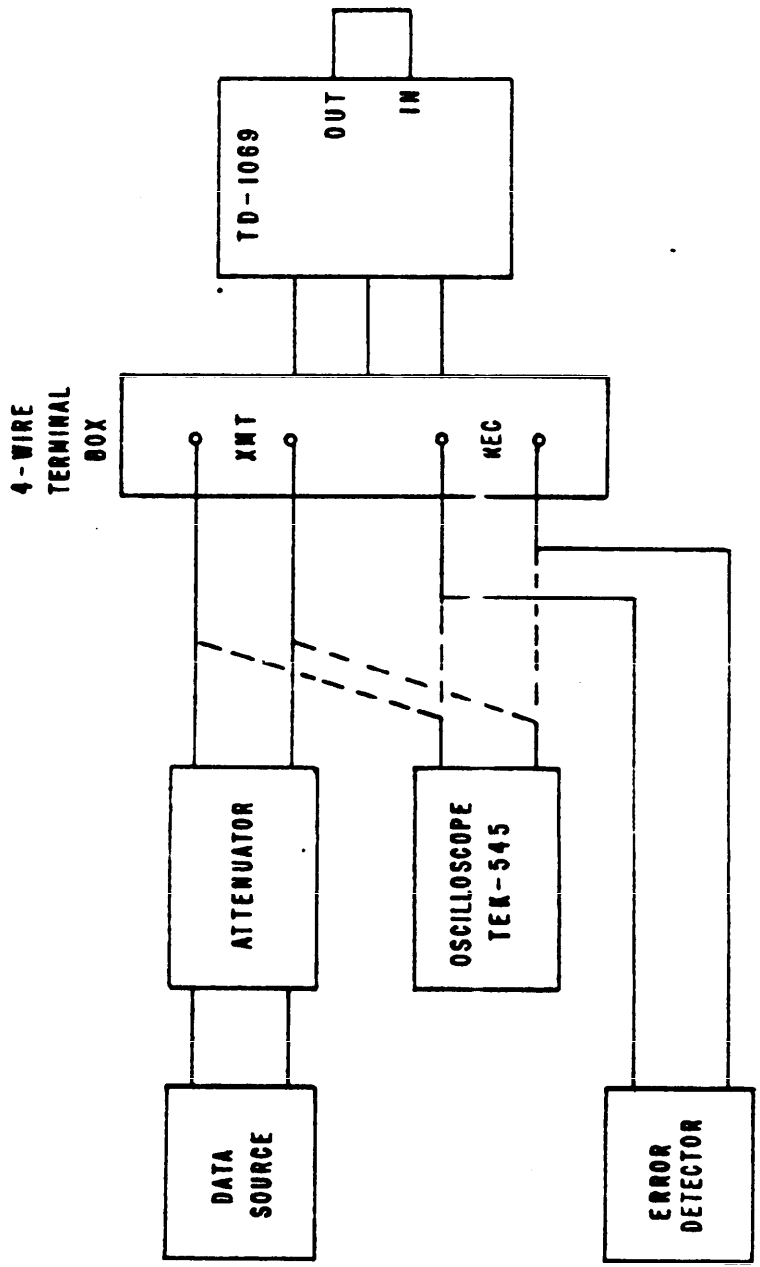


FIG. 3b CHANNEL INPUT SENSITIVITY AND OUTPUT VOLTAGE

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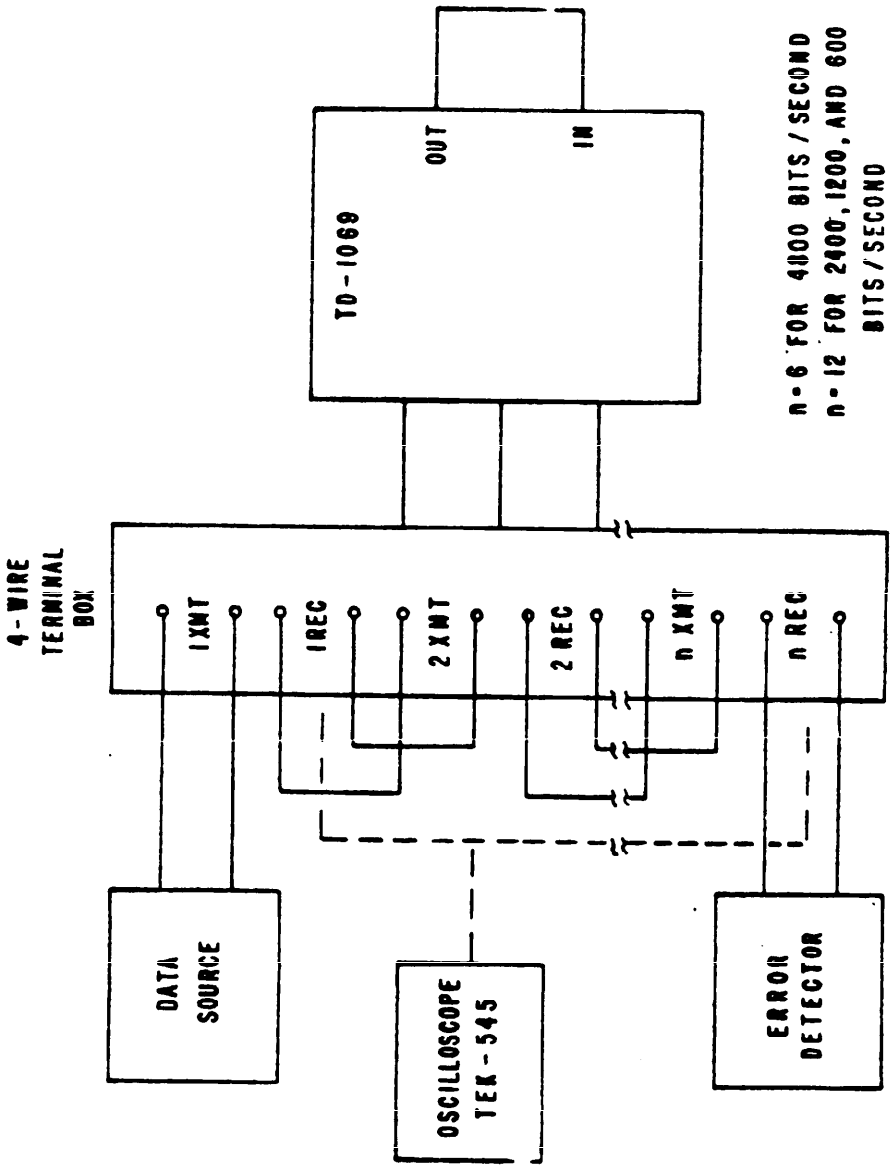


FIG. 3c CHANNEL WAVESHAPE AND RATE PERFORMANCE

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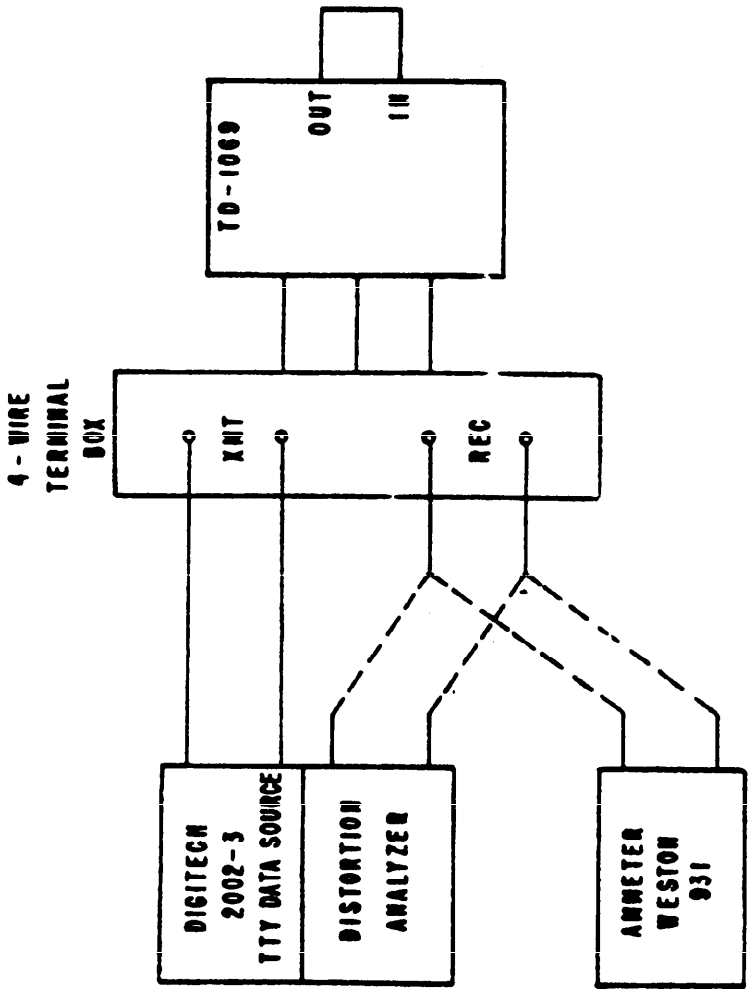


FIG. 3d TELETYPEWRITER CHANNEL PERFORMANCE

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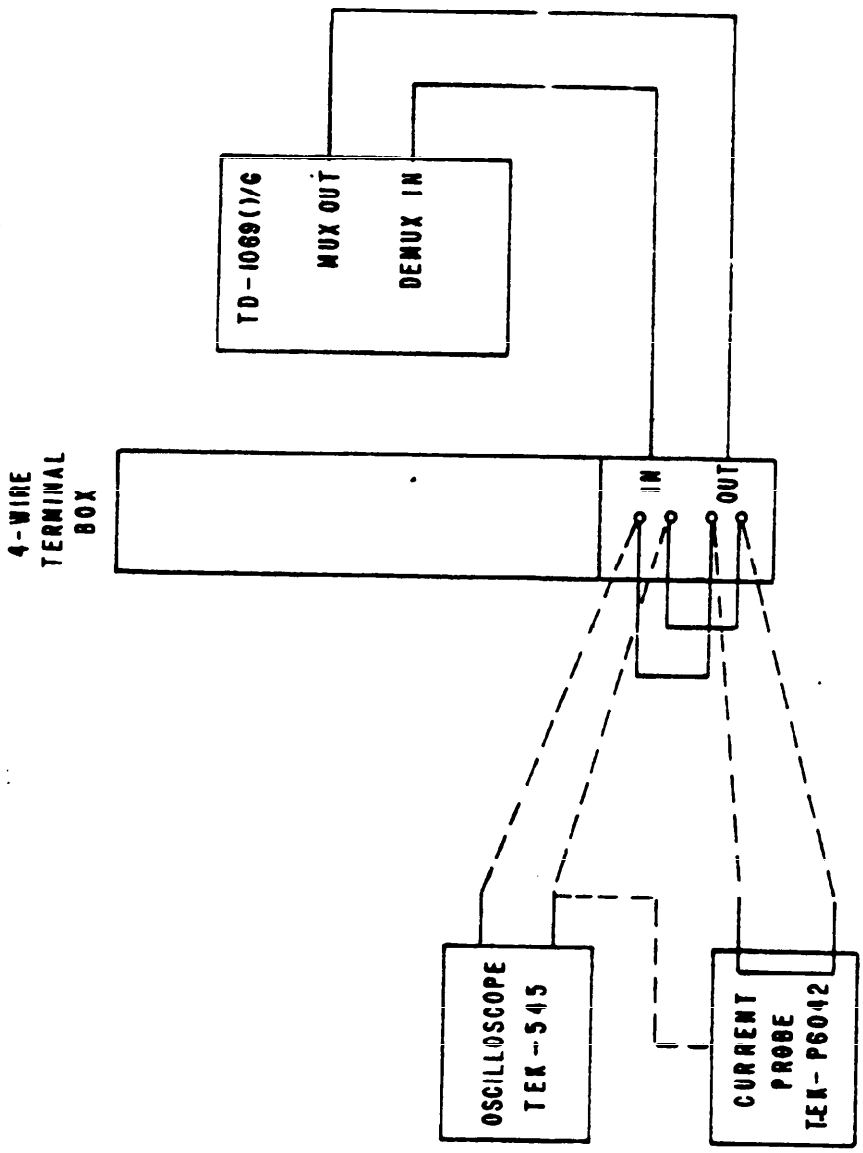


FIG. 3e MULTIPLEXER OUTPUT SIGNAL LEVEL

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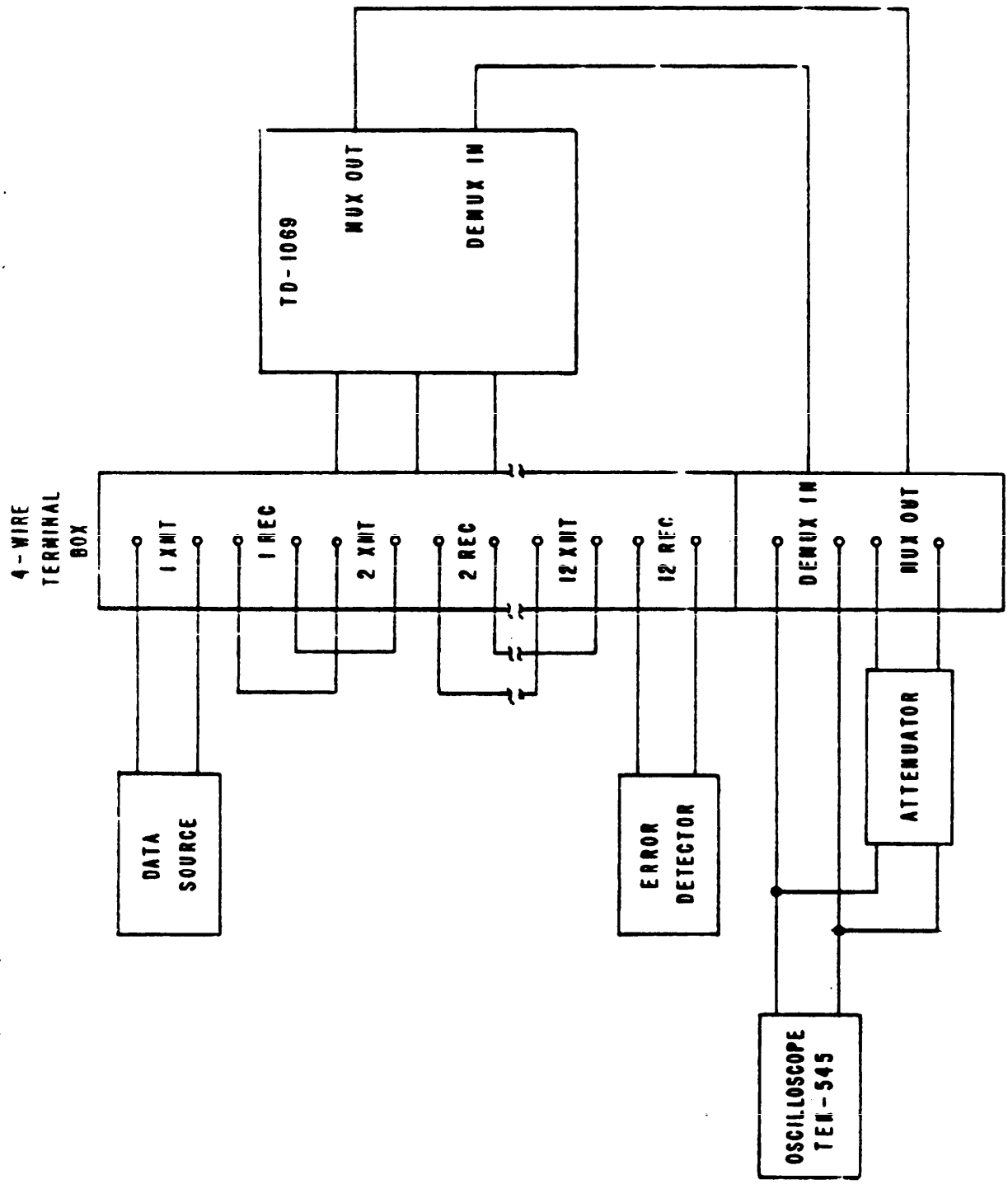


FIG.3f DEMULTIPLEXER INPUT SIGNAL LEVEL

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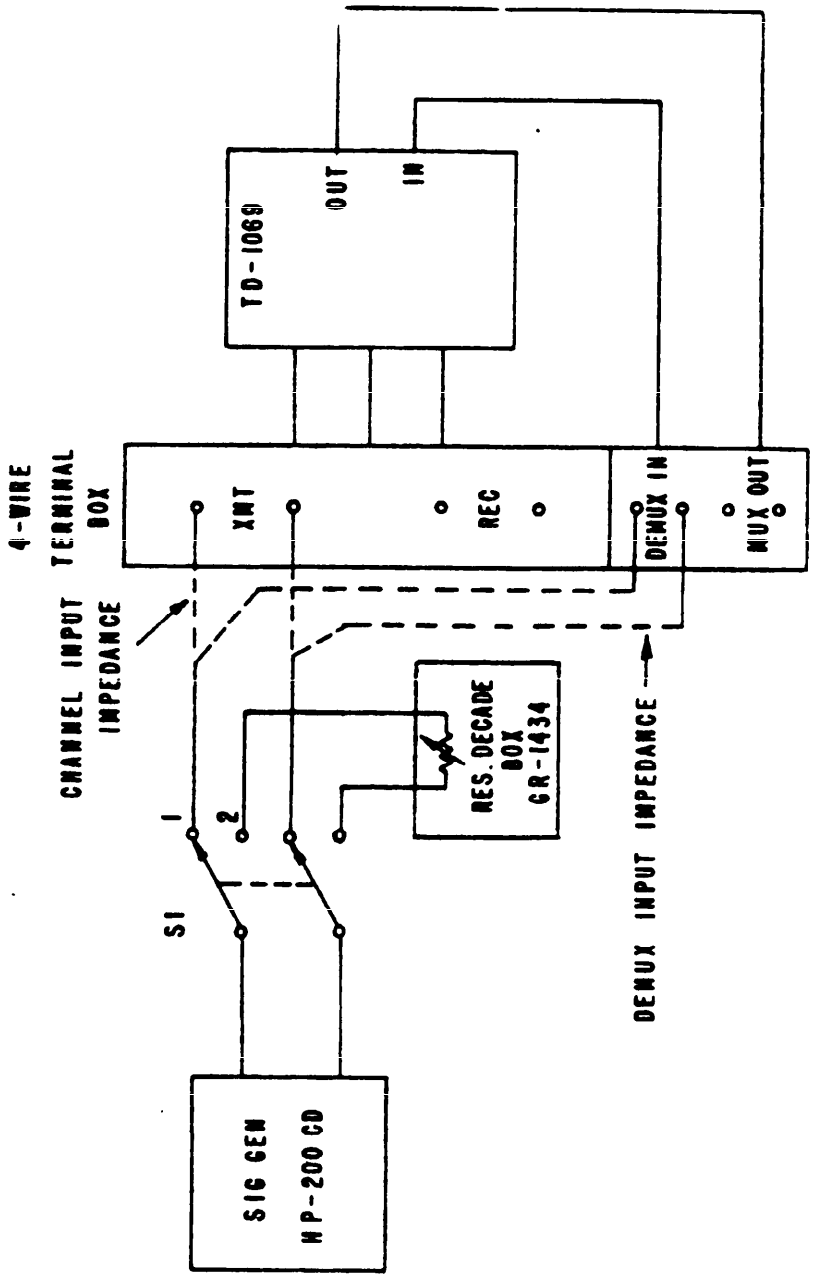


FIG. 4 INPUT IMPEDANCE

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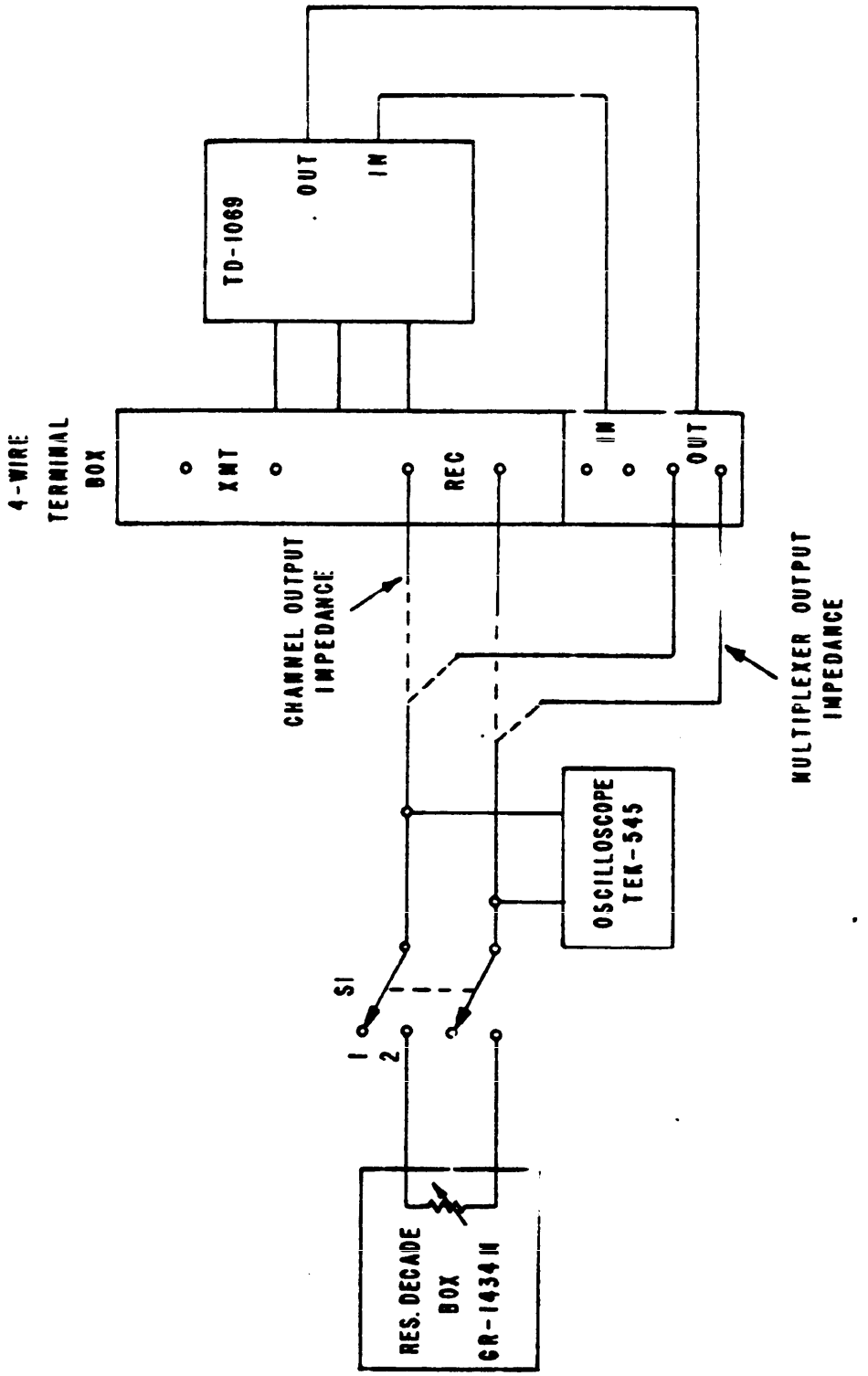


FIG.5 OUTPUT IMPEDANCE

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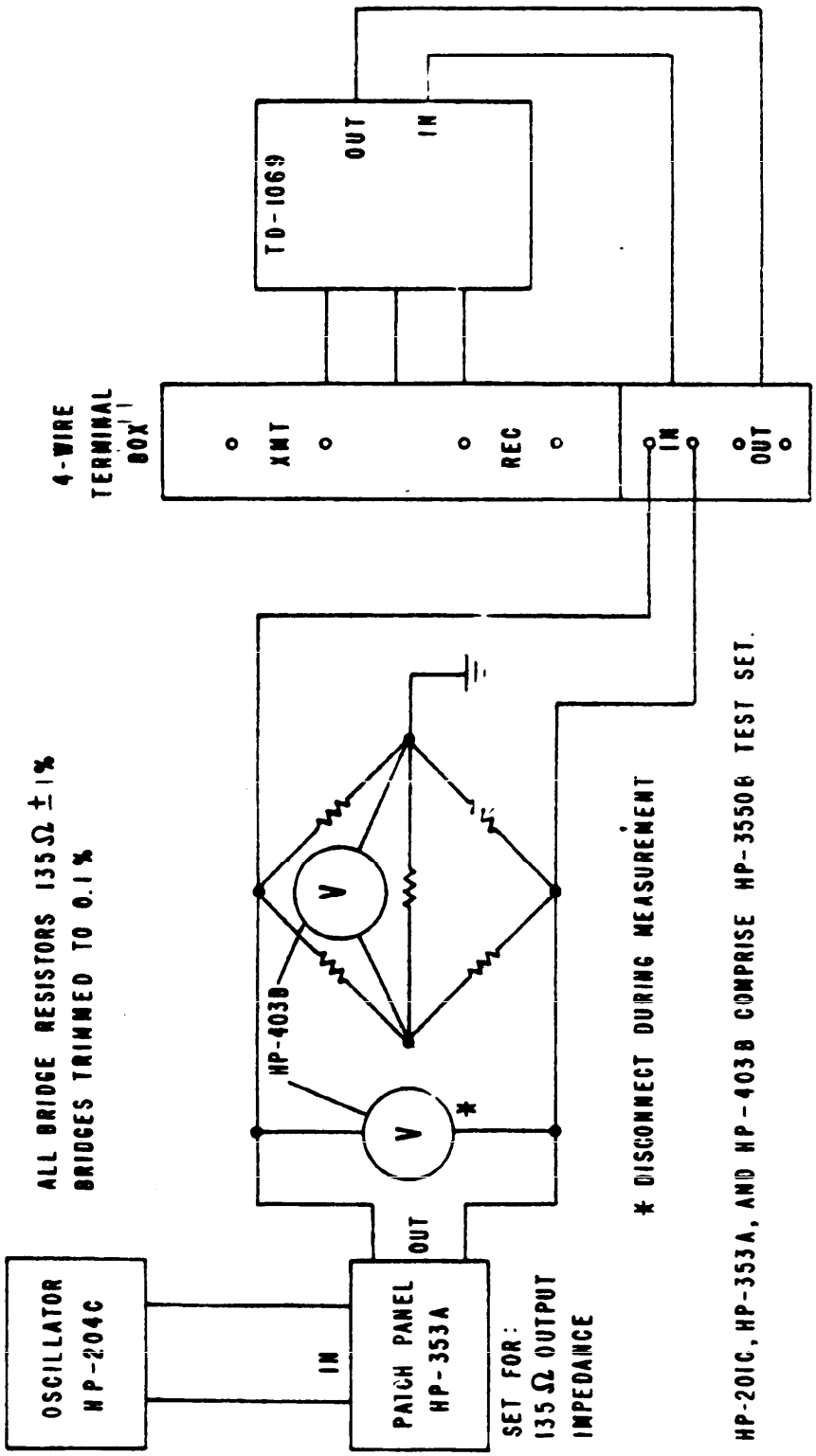


FIG. 6 MULTIPLEXER LONGITUDINAL BALANCE

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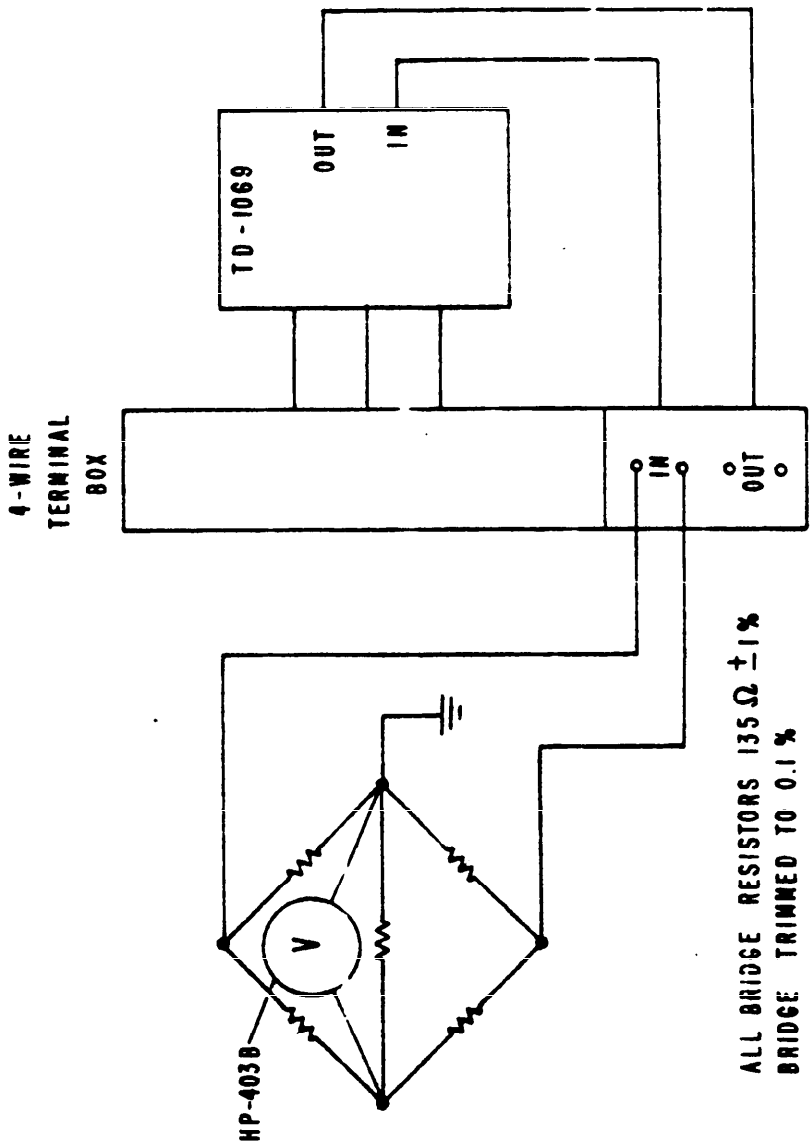


FIG. 7 DEMULTIPLEXER LONGITUDINAL BALANCE

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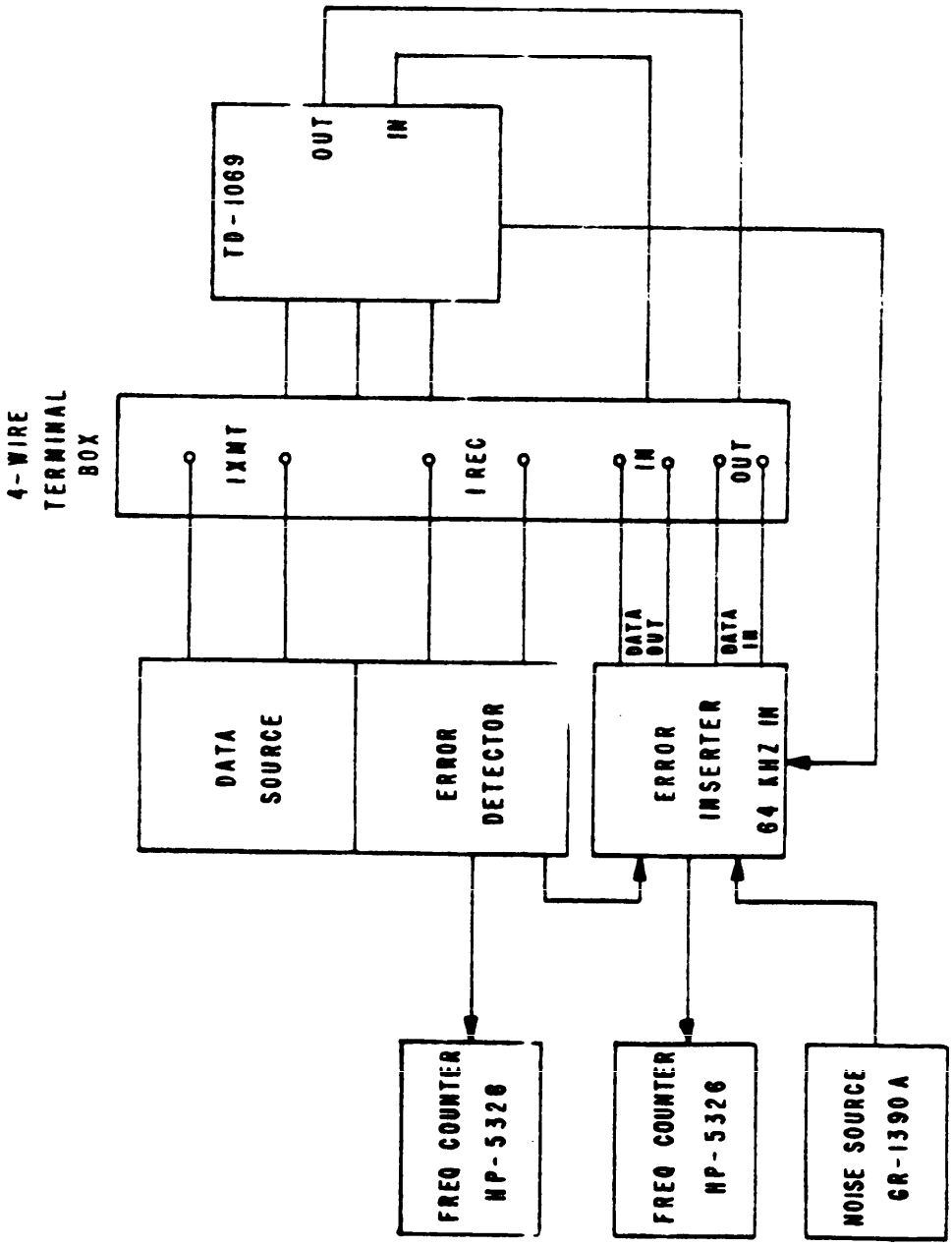


FIG. 8 BIT COUNT INTEGRITY

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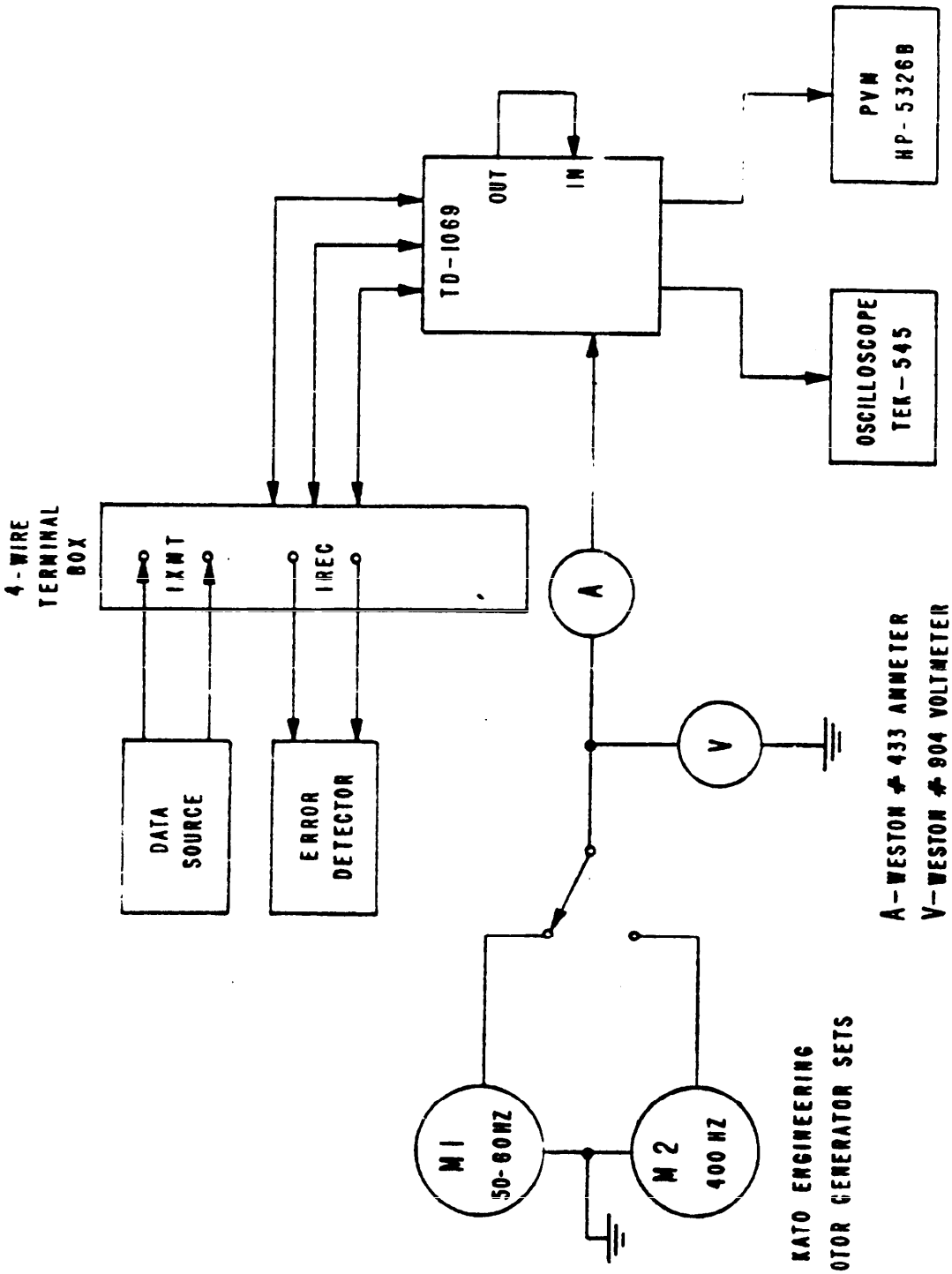


FIG.9 POWER SUPPLY TEST SET-UP

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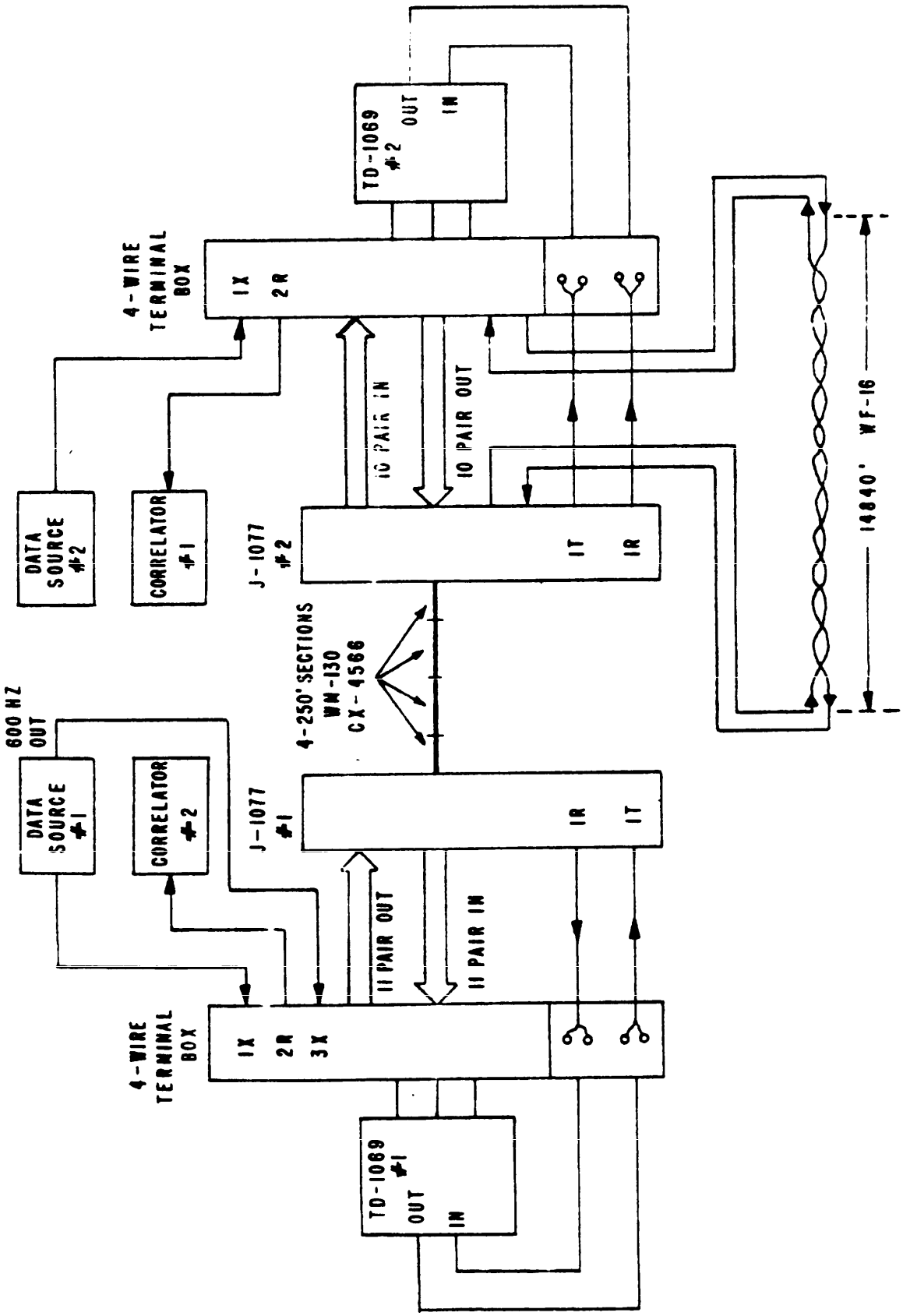


FIG. 10a INTERFACE CHARACTERISTICS

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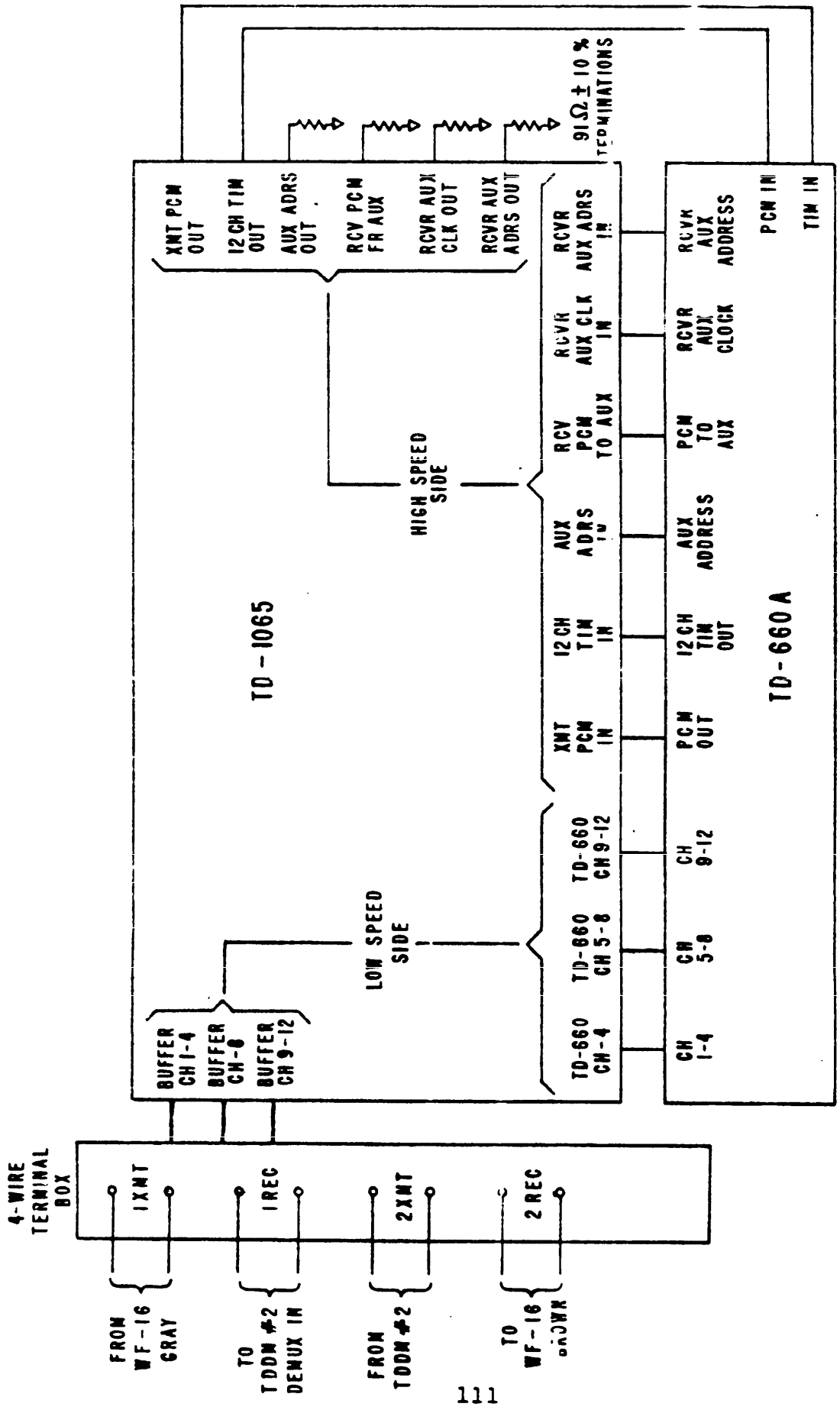


FIG. 10b TD-1069/TD-1065 COMPATABILITY

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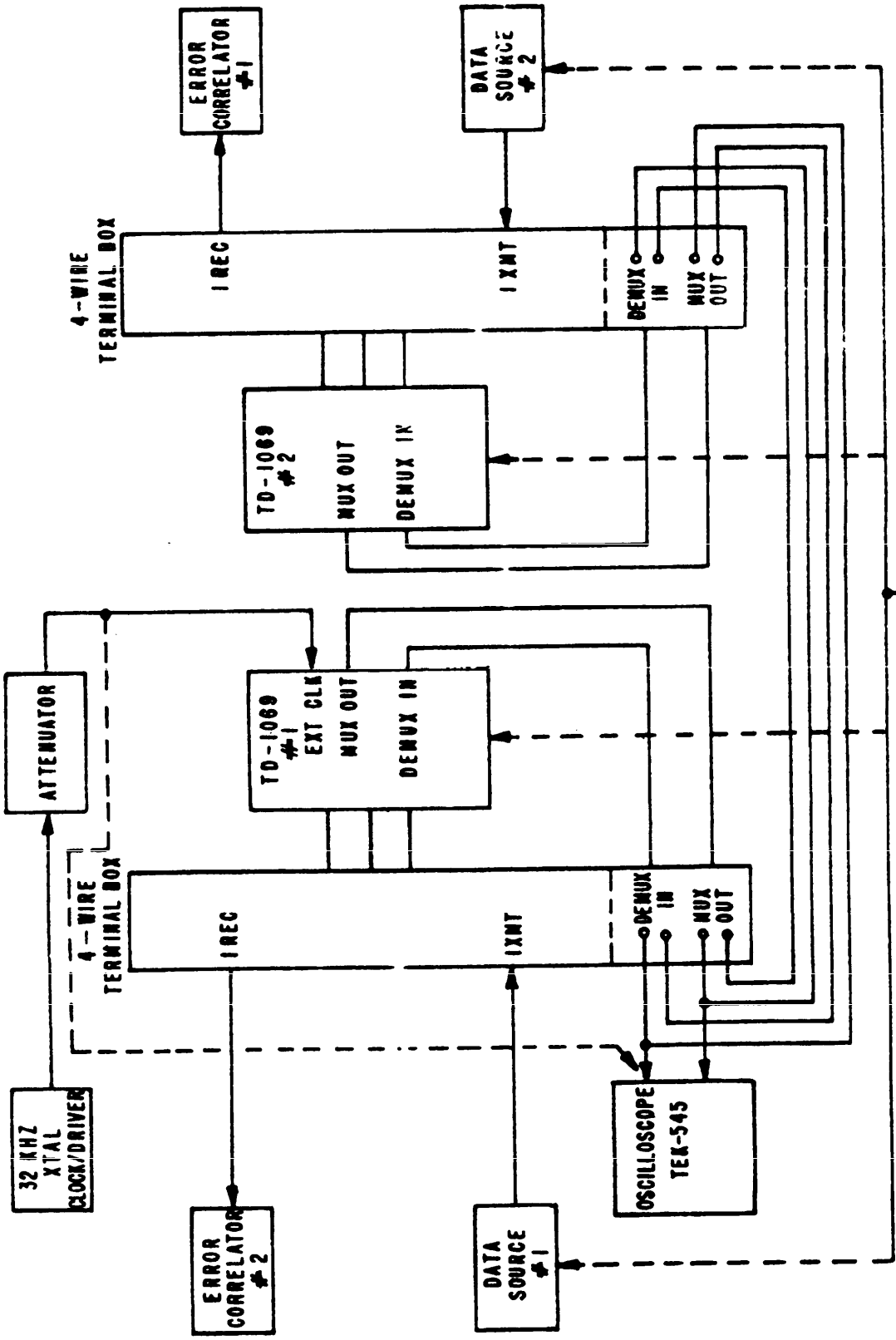


FIG. 11 INTERUNIT OPERATING CHARACTERISTICS

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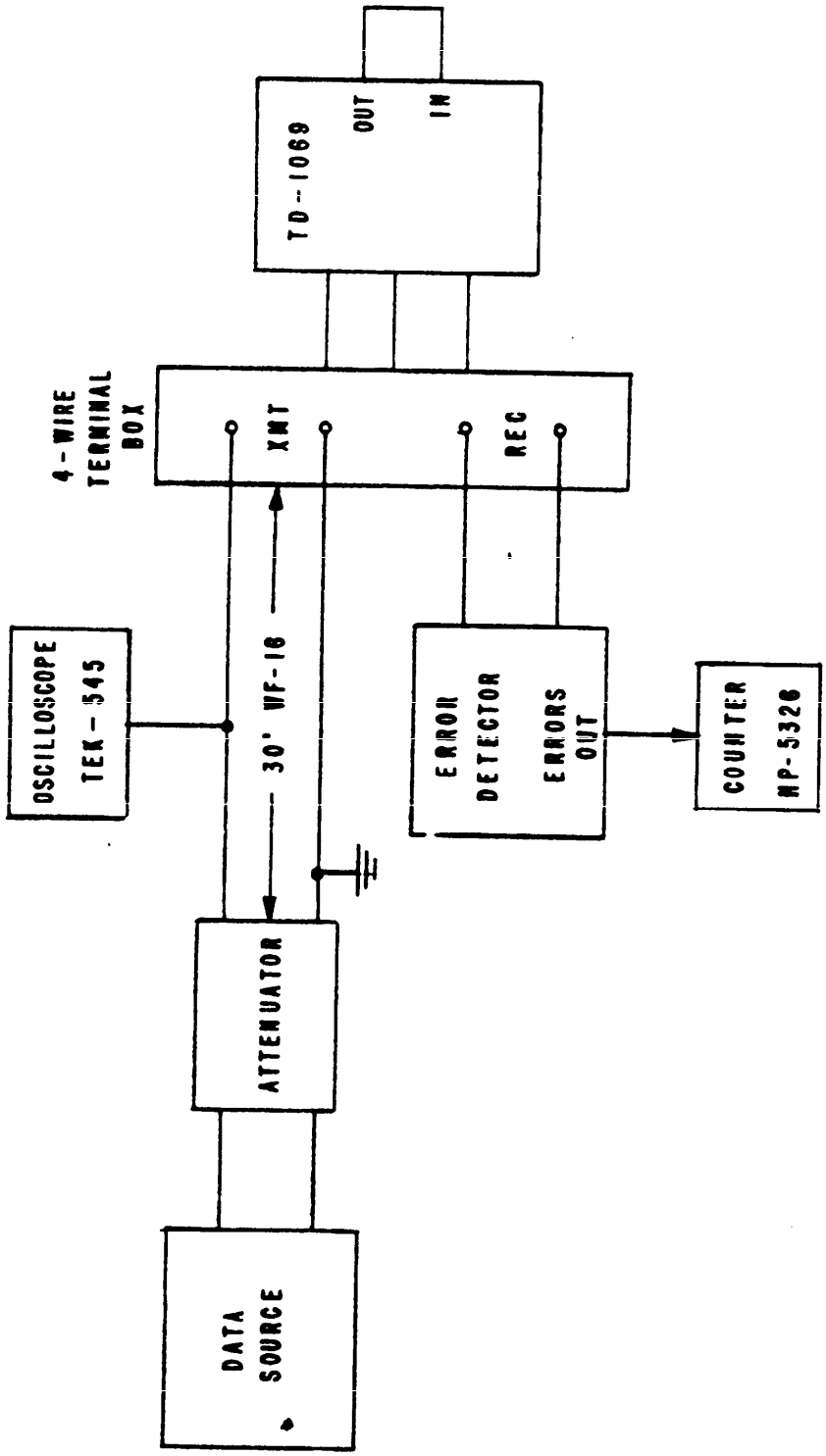


FIG. 12 UNBALANCED SIGNAL CHANNEL SENSITIVITY

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b. ADDRESS *(Street, City, State ZIP Code)*	☐ MANUFACTURER
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a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

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